Ordinary Differential Equation, Dynamical System, and Control Theory

沈威宇

September 12, 2026

Contents

Chapter 1 Ordinary Differential Equation (常微分方程), Dynamical System (動態系統 or 動力系統), and Control Theory (控制理論)

1 Ordinary Differential Equation

I Definition

i Ordinary Differential Equation (ODE) and System of ODEs (常微分方程組)

An nth-order ODE or system of ODEs of a unknown function or dependent variable unknown functions or dependent variables $\mathbf{y} : \mathbb{R} \to \mathbb{R}^m$ with respect to an independent variable $x \in \mathbb{R}$ can be written in the form, called normal form,

$$\mathbf{y}^{(n)} = F(x, mby, mby', mby'', \dots, \mathbf{y}^{(n-1)}),$$

in which $F : \Omega \subseteq \mathbb{R} \times \mathbb{R}^{m \times n} \to \mathbb{R}^m$ is a function.

ii Initial-value problem (IVP) (初值問題)

An nth-order initial-value problem is to solve an nth-order ODE or system of ODEs

$$\mathbf{y}^{(n)} = F(x, mby, mby', mby'', \dots, \mathbf{y}^{(n-1)})$$

that subjects to an condition called initial condition (IC) (初始條件) or conditions called initial conditions at a point $x_0 \in \mathbb{R}$ that can be written in the form

$$\mathbf{y}^{(i)}(x_0) = y_i, \quad \forall i \in \mathbb{N}_{0,<n}$$

and such that $(t, y_0, \dots, y_{n-1}) \in \Omega$.

A function $\mathbf{y} : \mathbb{R} \to \mathbb{R}^m$ that complies with the system of ODEs and the ICs is called a solution or solution vector.

The interval of existence of an IVP is the largest open interval containing x_0 where a solution of the IVP exists.

The interval of uniqueness of an IVP is the largest open interval containing x_0 where a unique solution of the IVP exists.

iii Boundary-value problem (BVP) (邊值問題)

An nth-order initial-value problem is to solve an nth-order ODE or system of ODEs that subjects to a condition called boundary condition (BC) (邊界條件) or conditions called boundary conditions.

A function $\mathbf{y}: \mathbb{R} \to \mathbb{R}^m$ that complies with the system of ODEs and the BCs is called a solution or solution vector.

II Solutions of ODEs

i Solution or flow of an ODE

Any function f that is defined on an interval I, called interval of definition, the interval of existence, the interval of validity, or the domain of the solution, and of class C^n on I, and when substituted into an nth-order ordinary differential equation reduces the equation to an identity, is said to be a solution (aka flow) of the equation on I.

An ODE does not necessarily have to possess a solution.

ii Solution curve or trajectory

The graph of a solution f on its interval of definition of an ODE is called a solution curve or a trajectory.

iii Explicit solution

An explicit solution of an ODE of a dependent variable y with respect to an independent variable x is in the form $y = f(x)$, where $f(x)$ is a function of x.

iv Implicit solution

A relation $G(x, y) = 0$ is said to be an implicit solution of an ODE on an interval I if there exists at least one function f that satisfes the relation $G(x, y) = 0$ as well as the ODE on I.

v Parametric family or parameterized family

A parametric family or a parameterized family is a family of objects whose differences depend only on the chosen values for a set of parameters.

vi Degrees of freedom

The degrees of freedom of a system is the number of parameters of the system that may vary independently.

vii Parametric family of solutions

An n-parameter family of solutions of an ODE of a dependent variable y with respect to an independent variable x is in the form $y = f(x, c_1, c_2, \dots c_n)$ (explicit) or $G(x, y, c_1, c_2, \dots c_n) = 0$

(implicit), in which $c_1, c_2, \dots c_n$ are parameters that are arbitrary given that the solution obtained is a solution of the ODE. The arbitrary parameters are usually denoted as $A, B, \dots$.

viii General solution (通解)

If every solution of an ODE on an interval I can be obtained from an n-parameter family of solutions by appropriate choices of the parameters, we then say that that family of solutions is the general solution of the ODE.

ix Particular solution (特解)

A particular solution of an ODE that is free of parameters is called a particular solution.

If a system given by an ODE have n degrees of freedom, one may obtain a determined particular solution by imposing $n - 1$ non contradictory constraints, leaving one degree of freedom determined by the ODE.

x Singular solution (奇異解)

Sometimes a differential equation possesses a solution that is not a member of a family of solutions of the equation, that is, a solution that cannot be obtained by specializing any of the parameters in the family of solutions. Such an extra solution is called a singular solution.

xi Solutions by substitutions or solutions by change of variables

When solving an ODE, we often replace dependent variables with functions of the independent variable and new dependent variables where the new dependent variables are functions of the independent variable and old dependent variables to transform it into another ODE of the new dependent variables that is easier to solve.

xii Solution of System of ODEs

A solution of a system of ODEs involving the derivatives of n unknown functions or dependent variables, $y_1, y_2, \dots y_n$ of a single independent variable x is a n-tuple of sufficiently smooth functions of x defined on a common interval of definition that satisfy all equations in the system.

A system of ODEs does not necessarily have to possess a solution.

xiii Orthogonal trajectory and isogonal trajectory

An orthogonal trajectory of a family of curves is a curve that intersects each curve of the family orthogonally.

An isogonal trajectory of a family of curves is a curve that intersects each curve of the family at a same angle α.

III Conversion of order to variable

Any nth-order system of ODEs of m variables $x_1, x_2, \dots, x_m$ can be converted to a first-order system of ODEs of $m \cdot n$ variables by, for each variable x, assigning $(n-1)$ new variables defined as the first to $(n-1)$th derivatives of x.

2 First-order ODE

Unless otherwise specified, the variables and functions are in $\mathbb{R}$.

I Differential form

The differential form of a first-order ODE $M(x,y) + N(x,y)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ is

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0.$$

II Slope

i Slope function or rate function

In a first-order ODE in the form $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x,y)$, f is called the slope function or rate function.

ii Direction field or slope field (斜率場)

In a first-order ODE in the form $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x,y)$, if we evaluate f over a rectangle grid of points and draw a lineal element at each point (x,y) of the grid with slope $f(x,y)$, then the collection of all these lineal elements is called a direction field or a slope field of the ODE.

iii Isocline

In a first-order ODE in the form $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x,y)$, the isocline for slope m is defined as the family of points $\{(x,y) \mid f(x,y) = m\}$. In the method of isocline, we draw a lineal element at each point in the isocline for slope m of f with slope m.

III Autonomous System

i Autonomous system (自治系統)

A system of nth-order ODEs is called autonomous if it can be written in the form

$$\frac{\mathrm{d}^n x}{\mathrm{d}t^n} = f(x, \frac{\mathrm{d}x}{\mathrm{d}t}, \dots, \frac{\mathrm{d}^{n-1}x}{\mathrm{d}t^{n-1}}).$$

ii Critical point (臨界點), equilibrium, equilibrium point (平衡點), or stationary point (駐點) of autonomous systems of ODEs

For an autonomous systems of ODEs, we say that a point c in the domain of f is a critical point, equilibrium, equilibrium point, or stationary point if $f(c) = 0$.

If c is a critical point, then $y(x) = c$ is a constant solution of the system of autonomous first-order ODEs, also called an equilibrium solution (平衡解).

If c is a critical point and f is locally Lipschitz at c, then $y(x) = c$ is the unique solution of the system of autonomous first-order ODEs through c.

iii Translation property or shift property of solutions of autonomous systems of ODEs

Let function $\Phi(t, y)$ denotes the solution of an autonomous system of ODEs with IC $x(0) = y$. The translation property or shift property states that

$$\forall s \in \mathbb{R}: \ \Phi(t + s, y) = \Phi(t, \Phi(s, y)).$$

IV Orthogonal trajectories

The family of the orthogonal trajectories of the family of solutions of the first-order ODE

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0,$$

is the family of solutions of the first-order ODE

$$N(x, y)\,\mathrm{d}x - M(x, y)\,\mathrm{d}y = 0.$$

V Separable equation

i Definition

A first-order ODE in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$$

is said to be separable or have separable variables.

ii Solutions by direct integration

The solutions of

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$$

are the family of solutions

$$\int \frac{1}{h(y)}\,\mathrm{d}y = \int g(x)\,\mathrm{d}x + C$$

and the constant solution

$$h(y) = 0.$$

iii Example: law of natural growth

IVP:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = kP, \quad P(0) = P_0.$$

Solutions:

$$P = P_0 e^{kt}.$$

Proof.

$$P^{-1}\,\mathrm{d}P = k\,\mathrm{d}t$$

$$\int P^{-1}\,\mathrm{d}P = \int k\,\mathrm{d}t = kt + A = \ln(P).$$

$$P = Ae^{kt}.$$

$$P_0 = A.$$

□

iv Example: logistic differential equation

IVP:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = kP(1 - \frac{P}{M}), \quad P(0) = P_0.$$

Solutions:

$$P = \frac{M}{1 + \frac{M-P_0}{P_0}e^{-kt}}.$$

Proof.

$$\frac{1}{P(1-\frac{P}{M})}\,\mathrm{d}P = k\,\mathrm{d}t$$

$$\int \frac{1}{P(1-\frac{P}{M})}\,\mathrm{d}P = \int k\,\mathrm{d}t = kt + A$$

$$\frac{1}{P(1-\frac{P}{M})} = \frac{M}{P(M-P)} = \frac{A}{P} + \frac{B}{M-P}$$

$$AM - AP + BP = M, \quad A = 1, \quad B = 1$$

$$\int \frac{1}{P(1-\frac{P}{M})}\,\mathrm{d}P = \int \frac{1}{P} + \frac{1}{M-P}\,\mathrm{d}P = \ln|P| - \ln|M-P| + C$$

$$\ln|P| - \ln|M-P| = kt + A = \ln\left|\frac{P}{M-P}\right|$$

$$\frac{P}{M-P} = Ae^{kt}$$

$$P = \frac{AMe^{kt}}{1 + Ae^{kt}}$$

$$P_0 = \frac{AM}{1+A}$$

$$A = \frac{P_0}{M-P_0}$$

$$P = \frac{\frac{P_0}{M-P_0}Me^{kt}}{1 + \frac{P_0}{M-P_0}e^{kt}} = \frac{M}{1 + \frac{M-P_0}{P_0}e^{-kt}}$$

□

v Example: Lotka–Volterra equations or predator–prey equations

ODE:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x(\alpha - \beta y),$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -y(\gamma - \delta x).$$

Implicit solutions:

$$\delta x - \gamma \ln(x) + \beta y - \alpha \ln(y) = C.$$

Equilibrium solutions:

$$x = 0, \quad y = 0,$$

and

$$x = \frac{\gamma}{\delta}, \quad y = \frac{\alpha}{\beta}.$$

Proof.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t}$$
$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{y}{x}\frac{\gamma - \delta x}{\alpha - \beta y}$$
$$\frac{\delta x - \gamma}{x}\,\mathrm{d}x = -\frac{\beta y - \alpha}{y}\,\mathrm{d}y$$
$$\delta x - \gamma \ln(x) + \beta y - \alpha \ln(y) = C$$

□

VI Linear equation

i Definition

A first-order ODE in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

is said to be linear.

ii Solutions by integrating factor

Use the integrating factor:

$$\mu(x) = e^{\int P(x)\mathrm{d}x}.$$

Multiply both sides with $\mu(x)$:

$$\mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)P(x)y = \mu(x)Q(x).$$

The left-hand side is exactly a derivative:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y(x)) = \mu(x)Q(x).$$

Integrate:

$$\mu(x)y(x) = \int \mu(x)Q(x)\,\mathrm{d}x + C.$$
$$y(x) = \frac{1}{\mu(x)}(\int \mu(x)Q(x)\,\mathrm{d}x + C).$$

iii Remarks

Occasionally, a first-order differential equation is not linear in one variable but is linear in the other variable. If a first-order ODE that can be wriiten in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{P(y)x + Q(y)}.$$

We can take its reciprocal as

$$\frac{\mathrm{d}x}{\mathrm{d}y} - P(y)x = Q(y),$$

which is recognized as linear in the variable x.

iv Example: constant voltage series RL filter circuit

IVP:

$$L\frac{\mathrm{d}I(t)}{\mathrm{d}t} + RI(t) = V, \quad I(0) = I_0.$$

Solutions:

$$I(t) = \frac{V}{R}(1 - e^{-Rt/L}) + I_0 e^{-Rt/L}.$$

Equilibrium solution:

$$I(t) = \frac{V}{R}.$$

Proof.

$$\frac{\mathrm{d}I(t)}{\mathrm{d}t} + \frac{R}{L}I(t) = \frac{V}{L}$$

$$\mu(t) = \exp\left(\int \frac{R}{L}\,\mathrm{d}t\right) = e^{Rt/L}$$

$$e^{Rt/L}\frac{\mathrm{d}I(t)}{\mathrm{d}t} + e^{Rt/L}\frac{R}{L}I(t) = \frac{\mathrm{d}u}{\mathrm{d}t}, \quad u = e^{Rt/L}I(t)$$

$$\int 1\,\mathrm{d}u = \int e^{Rt/L}\frac{V}{L}\,\mathrm{d}t = u + A = \frac{V}{R}e^{Rt/L} + B$$

$$u = e^{Rt/L}I(t) = \frac{V}{R}e^{Rt/L} + A$$

$$I(t) = \frac{V}{R} + Ae^{-Rt/L}$$

$$I_0 = \frac{V}{R} + A$$

$$A = I_0 - \frac{V}{R}$$

$$I(t) = \frac{V}{R}(1 - e^{-Rt/L}) + I_0 e^{-Rt/L}$$

□

v Example: constant current parallel RC filter circuit

IVP:

$$C\frac{\mathrm{d}V(t)}{\mathrm{d}t} + \frac{1}{R}V(t) = I, \quad V(0) = V_0.$$

Solutions:

$$V(t) = IR(1 - e^{-t/(RC)}) + V_0 e^{-t/(RC)}.$$

Equilibrium solution:

$$V(t) = IR.$$

Proof.

$$\frac{\mathrm{d}V(t)}{\mathrm{d}t} + \frac{1}{RC}V(t) = \frac{I}{C}$$

$$\mu(t) = \exp\left(\int \frac{1}{RC}\,\mathrm{d}t\right) = e^{t/(RC)}$$

$$e^{t/(RC)}\frac{\mathrm{d}V(t)}{\mathrm{d}t} + e^{t/(RC)}\frac{1}{RC}V(t) = e^{t/(RC)}\frac{I}{C} = \frac{\mathrm{d}u}{\mathrm{d}t}, \quad u = e^{t/(RC)}V(t)$$

$$\int 1\,\mathrm{d}u = \int e^{t/(RC)}\frac{I}{C}\,\mathrm{d}t = u + A = IRe^{t/(RC)} + B$$

$$u = e^{t/(RC)}V(t) = IRe^{t/(RC)} + A$$

$$V(t) = IR + Ae^{-t/(RC)}$$

$$V_0 = IR + A$$

$$A = V_0 - IR$$

$$V(t) = IR(1 - e^{-t/(RC)}) + V_0 e^{-t/(RC)}$$

□

vi Example: Falling with drag force complying with Stokes' law

IVP:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -kv - mg, \quad v(0) = v_0.$$

Solutions:

$$v = \frac{mg}{k}(1 - e^{-kt/m}) + v_0 e^{-kt/m}.$$

Equilibrium solution:

$$v = \frac{mg}{k}.$$

Proof.

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{k}{m}v = -g$$

$$\mu(t) = \exp\left(\int \frac{k}{m}\,\mathrm{d}t\right) = e^{kt/m}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(e^{kt/m}v) = -e^{kt/m}g$$

$$e^{kt/m}v = \int -e^{kt/m}g\,\mathrm{d}t = \frac{mg}{k}e^{kt/m} + C$$

$$v = \frac{mg}{k} + Ce^{-kt/m}$$
$$v_0 = \frac{mg}{k} + C$$
$$v = \frac{mg}{k} + (v_0 - \frac{mg}{k})e^{-kt/m}.$$

□

VII Exact equation

i Definition

A first-order ODE in the form

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

where $\frac{\partial^2 M}{\partial x \partial y}$ and $\frac{\partial^2 N}{\partial x \partial y}$ is continuous and

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

is said to be exact, and its LHS is called exact differential.

By Schwarz's Theorem, the criterion holds iff there exists a function $F : R \to \mathbb{R}$ such that

$$\frac{\partial F}{\partial x} = M, \quad \frac{\partial F}{\partial y} = N,$$

and $\frac{\partial^2 F}{\partial x \partial y}$ is continuous. Such F is called the potential function.

ii Solutions by integration and then differentiation with respect to different variables

First, integrate $M(x, y)$ with respect to x:

$$F(x, y) = \int M(x, y)\,\mathrm{d}x + \phi(y),$$

where $\phi(y)$ is an arbitrary constant of integration that may depend on y.

Second, differentiate with respect to y:

$$\frac{\partial F}{\partial y}(x, y) = N(x, y) = \frac{\partial}{\partial y}(\int M(x, y)\,\mathrm{d}x) + \phi'(y).$$

Third, integrate $\phi'(y)$ with respect to y to find $\phi(y)$:

$$\phi(y) = \int \phi'(y)\,\mathrm{d}y = \int N(x, y) - \frac{\partial}{\partial y}(\int M(x, y)\,\mathrm{d}x)\,\mathrm{d}y.$$

Fourth, substitute $\phi(y)$ back to find the potential function $F(x, y)$. The implicit solutions is given by

$$F(x, y) = C,$$

where C is an arbitrary constant.

Alternatively, first, integrate $N(x, y)$ with respect to y:

$$F(x, y) = \int N(x, y)\,\mathrm{d}y + \phi(x),$$

where $\phi(x)$ is an arbitrary constant of integration that may depend on x.

Second, differentiate with respect to x:

$$\frac{\partial F}{\partial x}(x, y) = M(x, y) = \frac{\partial}{\partial x}\left(\int N(x, y)\,\mathrm{d}y\right) + \phi'(x).$$

Third, integrate $\phi(x)$ with respect to x to find $\phi(x)$:

$$\phi(x) = \int \phi'(x)\,\mathrm{d}x = \int M(x, y) - \frac{\partial}{\partial x}\left(\int N(x, y)\,\mathrm{d}y\right)\mathrm{d}x\,.$$

Fourth, substitute $\phi(x)$ back to find the potential function $F(x, y)$. The implicit solutions is given by

$$F(x, y) = C,$$

where C is an arbitrary constant.

iii **Example:** $(4x^3 + 3y^2 + \cos(x))\,\mathrm{d}x + (6xy + 2)\,\mathrm{d}y = 0$

ODE:

$$(4x^3 + 3y^2 + \cos(x))\,\mathrm{d}x + (6xy + 2)\,\mathrm{d}y = 0.$$

Solutions:

$$x^4 + 3xy^2 + \sin(x) + 2y = C.$$

Proof.

$$\begin{aligned}
\frac{\partial}{\partial y}\left(4x^3 + 3y^2 + \cos(x)\right) &= 6y \\
\frac{\partial}{\partial x}(6xy + 2) &= 6y \\
F(x, y) &= \int 4x^3 + 3y^2 + \cos(x)\,\mathrm{d}x + \phi(y) \\
&= x^4 + 3xy^2 + \sin(x) + \phi(y) \\
\frac{\partial F}{\partial y} &= 6xy + \phi'(y) = 6xy + 2 \\
\phi'(y) = 2, &\quad \phi(y) = 2y + C \\
F(x, y) &= x^4 + 3xy^2 + \sin(x) + 2y + C
\end{aligned}$$

□

VIII Integration factors to make inexact equations exact

i General case

Given a first-order ODE in the form

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0,$$

we define the integration factor $\mu(x, y)$ such that

$$\mu(x, y)M(x, y)\,\mathrm{d}x + \mu(x, y)N(x, y)\,\mathrm{d}y = 0$$

is exact and with potential function F such that

$$\frac{\partial F}{\partial x} = \mu(x, y)M(x, y), \quad \frac{\partial F}{\partial y} = \mu(x, y)N(x, y),$$

that is,

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x}.$$

Solve the PDE to get μ.

ii Separable case

If

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

depends only on x, we can construct integration factor $\mu(x)$:

$$\mu(x) = e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}\mathrm{d}x}$$

such that

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x} = \mu\frac{\partial M}{\partial y}.$$

Similarly, if

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

depends only on y, we can construct integration factor $\mu(y)$:

$$\mu(y) = e^{\int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}\mathrm{d}x}$$

such that

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x} = \mu\frac{\partial N}{\partial x}.$$

Proof.

$$\frac{\mu'(x)}{\mu(x)} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}.$$

This is a separable ODE of μ with repect to x.

Let

$$P(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}.$$

$$\mu^{-1}\,\mathrm{d}\mu = P(x)\,\mathrm{d}x\,.$$

$$\ln|\mu| = \int P(x)\,\mathrm{d}x\,.$$

$$\mu = e^{\int P(x)\mathrm{d}x}.$$

Similarly,

$$\frac{\mu'(y)}{\mu(y)} = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

for the second case. □

IX Example: $(5xy + 4y^2 + 1)\,\mathrm{d}x + (x^2 + 2xy)\,\mathrm{d}y = 0$

ODE:

$$(5xy + 4y^2 + 1)\,\mathrm{d}x + (x^2 + 2xy)\,\mathrm{d}y = 0.$$

Solutions:

$$x^5 y + x^4 y^2 + \frac{x^4}{4} = C.$$

Proof.

$$\frac{\partial}{\partial y}\big(5xy + 4y^2 + 1\big) = 5x + 8y$$

$$\frac{\partial}{\partial x}\big(x^2 + 2xy\big) = 2x + 2y$$

$$\frac{\frac{\partial}{\partial y}\big(5xy + 4y^2 + 1\big) - \frac{\partial}{\partial x}\big(x^2 + 2xy\big)}{x^2 + 2xy} = \frac{3x + 6y}{x^2 + 2xy} = \frac{3}{x}$$

$$\mu(x) = \exp\left(\int \frac{3}{x}\,\mathrm{d}x\right) = x^3$$

$$x^3(5xy + 4y^2 + 1)\,\mathrm{d}x + x^3(x^2 + 2xy)\,\mathrm{d}y = 0$$

$$\begin{aligned} F(x, y) &= \int x^3(5xy + 4y^2 + 1)\,\mathrm{d}x + \phi(y) \\ &= \int 5x^4 y + 4x^3 y^2 + x^3\,\mathrm{d}x + \phi(y) \\ &= x^5 y + x^4 y^2 + \frac{x^4}{4} + \phi(y) \end{aligned}$$

$$\frac{\partial F}{\partial y} = x^5 + 2x^4 y + \phi'(y) = x^3(x^2 + 2xy) = x^5 + 2x^4 y$$

$$\phi'(y) = 0, \quad \phi(y) = C$$

$$F(x, y) = x^5 y + x^4 y^2 + \frac{x^4}{4} + C$$

□

i Example: $(2xy+y)\,\mathrm{d}x + x^2\,\mathrm{d}y = 0$

ODE:

$$(2xy+y)\,\mathrm{d}x + x^2\,\mathrm{d}y = 0.$$

Solutions:

$$y = Ce^{1/x}x^{-2}.$$

Proof.

$$\frac{\partial}{\partial y}(2xy+y) = 2x+1$$

$$\frac{\partial}{\partial x}\left(x^2\right) = 2x$$

$$\frac{\frac{\partial}{\partial y}(2xy+y) - \frac{\partial}{\partial x}\left(x^2\right)}{x^2} = \frac{1}{x^2}$$

$$\mu(x) = \exp\left(\int \frac{1}{x^2}\,\mathrm{d}x\right) = e^{-1/x}$$

$$e^{-1/x}(2xy+y)\,\mathrm{d}x + e^{-1/x}x^2\,\mathrm{d}y = 0.$$

$$F(x,y) = \int e^{-1/x}x^2\,\mathrm{d}y + \phi(x) = e^{-1/x}x^2y + \phi(x)$$

$$\frac{\partial F}{\partial x} = (x^{-2}e^{-1/x}x^2 + e^{-1/x}2x)y + \phi'(x) = e^{-1/x}(2x+1)y + \phi'(x)$$

$$\phi'(x) = 0$$

$$F(x,y) = e^{-1/x}x^2y + C$$

$$e^{-1/x}x^2y = C$$

$$y = Ce^{1/x}x^{-2}$$

□

X Homogeneous equation

i Definition

A first-order ODE in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = F(x,y)$$

with $F(x,y)$ be a homogeneous function of degree 0, i.e., $F(tx.ty) = F(x,y)$, is said to be homogeneous.

A first-order ODE in the form

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0,$$

where $M(x,y)$ and $N(x,y)$ are homogeneous functions of the same degree is homogeneous since $F(x,y) = -\dfrac{M(x,y)}{N(x,y)}$ is a homogeneous function of degree 0.

ii Solutions by substitutions

For any homogeneous function $F(x, y)$ of 2 variables, there exists a function f of 1 variable such that $F(x, y) = f(\frac{y}{x})$ for all $x \neq 0$ such that (x, y) is in the domain of F.

Introduce the substitution $v = \frac{y}{x}$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}(xv)}{\mathrm{d}x} = v + x\frac{\mathrm{d}v}{\mathrm{d}x} = f(v),$$

which is a separable ODE.

$$\frac{\mathrm{d}v}{f(v) - v} = \frac{\mathrm{d}x}{x}.$$

$$\int \frac{\mathrm{d}v}{f(v) - v} = \ln|x| + C.$$

Substituting $v = \frac{y}{x}$ back gives the solutions.

If F is defined when $x = 0$, we also have a implicit constant solution

$$F(0, y) = 0.$$

iii **Example:** $(x^2 + y^2)\,\mathrm{d}x - 2xy\,\mathrm{d}y = 0$

ODE:

$$(x^2 + y^2)\,\mathrm{d}x - 2xy\,\mathrm{d}y = 0.$$

Implicit solutions:

$$y^2 = x^2 + Cx.$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 + y^2}{2xy} &= F(x, y) \\
F(tx, ty) &= F(x, y) \\
v &= \frac{y}{x} \\
F(x, y) = f(v) &= \frac{1 + v^2}{2v} \\
\frac{\mathrm{d}y}{\mathrm{d}x} = v + x\frac{\mathrm{d}v}{\mathrm{d}x} &= f(v) \\
\frac{1}{f(v) - v}\,\mathrm{d}v &= \frac{\mathrm{d}x}{x} \\
\int \frac{1}{f(v) - v}\,\mathrm{d}v = \ln|x| + C & \\
&= \int \frac{2v}{1 - v^2}\,\mathrm{d}v \\
&= -\ln|u|, \quad u = 1 - v^2 \\
&= -\ln\left|1 - v^2\right| \\
C|x| &= \frac{1}{|1 - v^2|}
\end{aligned}$$

$$v^2 = 1 + \frac{C}{x}$$
$$y^2 = x^2 + Cx$$

□

iv Example: $(x^2 + y^2)\,\mathrm{d}x + y^2\,\mathrm{d}y = 0$

ODE:

$$(x^2 + y^2)\,\mathrm{d}x + y^2\,\mathrm{d}y = 0.$$

Implicit solutions:

$$\arctan\left(\frac{2\sqrt{3}}{3}(\frac{y}{x} + \frac{1}{2})\right) = \frac{\sqrt{3}}{2}\ln\left|\frac{x^2}{y}\right| + C.$$

Constant solution:

$$x = 0, \quad y = 0.$$

Proof.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{y^2}{x^2 + y^2} = F(x, y)$$
$$F(tx, ty) = F(x, y)$$
$$v = \frac{y}{x}$$
$$F(x, y) = f(v) = -\frac{v^2}{1 + v^2}, \quad x \neq 0$$
$$\frac{1}{f(v) - v}\,\mathrm{d}v = \frac{\mathrm{d}x}{x}$$
$$\int \frac{1}{f(v) - v}\,\mathrm{d}v = \ln|x| + C$$
$$\frac{1}{f(v) - v} = -\frac{1 + v^2}{v(1 + v + v^2)} = -\frac{A}{v} - \frac{Bv + C}{1 + v + v^2}$$
$$A + Av + Av^2 + Bv^2 + Cv = 1 + v^2$$
$$A = 1, \quad B = 0, \quad C = -1$$
$$\begin{aligned}
\int \frac{1}{f(v) - v}\,\mathrm{d}v &= -\int \frac{1}{v}\,\mathrm{d}v + \int \frac{1}{1 + v + v^2}\,\mathrm{d}v \\
&= \ln|v| + \int \frac{1}{(v + \frac{1}{2})^2 + \frac{3}{4}}\,\mathrm{d}v, \quad u = \frac{2}{\sqrt{3}}(v + \frac{1}{2}) \\
&= \ln|v| + \frac{\sqrt{3}}{2}\frac{4}{3}\int \frac{1}{u^2 + 1}\,\mathrm{d}u \\
&= \ln|v| + \frac{2\sqrt{3}}{3}\arctan(u) + C \\
&= \ln|v| + \frac{2\sqrt{3}}{3}\arctan\left(\frac{2\sqrt{3}}{3}(v + \frac{1}{2})\right) + C
\end{aligned}$$
$$\ln|v| + \frac{2\sqrt{3}}{3}\arctan\left(\frac{2\sqrt{3}}{3}(v + \frac{1}{2})\right) = \ln|x| + C$$

$$\ln|y| - \ln|x| + \frac{2\sqrt{3}}{3}\arctan\left(\frac{2\sqrt{3}}{3}(\frac{y}{x}+\frac{1}{2})\right) = \ln|x| + C$$

$$\arctan\left(\frac{2\sqrt{3}}{3}(\frac{y}{x}+\frac{1}{2})\right) = \frac{\sqrt{3}}{2}\ln\left|\frac{x^2}{y}\right| + C$$

$$\arctan\left(\frac{2\sqrt{3}}{3}(\frac{y}{x}+\frac{1}{2})\right) = \frac{\sqrt{3}}{2}\ln\left|\frac{x^2}{y}\right| + C$$

$$F(0, y) = 0, \quad y = 0$$

□

XI Bernoulli equation

i Definition

A first-order ODE in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)y^n, \quad n \in \mathbb{R} \setminus \{0, 1\}$$

is called a Bernoulli equation.

When $n = 0$ or 1, it is a first-order linear equation.

ii Solutions by substitutions

Divide both sides by y^n:

$$y^{-n}\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y^{1-n} = Q(x).$$

Introduce the substitution $v = y^{1-n}$.

$$\frac{\mathrm{d}v}{\mathrm{d}x} = (1-n)y^{-n}\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Substitute into the equation:

$$\frac{1}{1-n}\frac{\mathrm{d}v}{\mathrm{d}x} + P(x)v = Q(x).$$

$$\frac{\mathrm{d}v}{\mathrm{d}x} + (1-n)P(x)v = (1-n)Q(x),$$

which is linear.

Use the integrating factor:

$$\mu(x) = e^{\int (1-n)P(x)\mathrm{d}x}.$$

Multiply both sides with $\mu(x)$:

$$\mu(x)\frac{\mathrm{d}v}{\mathrm{d}x} + \mu(x)(1-n)P(x)v = \mu(x)(1-n)Q(x).$$

The left-hand side is exactly a derivative:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)v(x)) = \mu(x)(1-n)Q(x).$$

Integrate:

$$\mu(x)v(x) = \int \mu(x)(1-n)Q(x)\,\mathrm{d}x + C.$$

$$v(x) = \frac{1}{\mu(x)}(\int \mu(x)(1-n)Q(x)\,\mathrm{d}x + C).$$

Recall $v = y^{1-n}$, we obtain the family of general solutions.

iii **Example:** $\frac{\mathrm{d}y}{\mathrm{d}x} + \cos(x)y = \sin(2x)y^{3/2}$

ODE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \cos(x)y = \sin(2x)y^{3/2}.$$

Solutions:

$$y = (2\sin(x) + 4 + Ce^{\sin(x)/2})^{-2}.$$

Proof.

$$y^{-3/2}\frac{\mathrm{d}y}{\mathrm{d}x} + \cos(x)y^{-1/2} = 2\sin(x)\cos(x)$$

$$v = y^{-1/2}, \quad \frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{1}{2}y^{-3/2}\frac{\mathrm{d}y}{\mathrm{d}x}.$$

$$\frac{\mathrm{d}v}{\mathrm{d}x} - \frac{1}{2}\cos(x)v = -\sin(x)\cos(x)$$

$$\mu(x) = \exp\left(\int -\frac{1}{2}\cos(x)\,\mathrm{d}x\right) = e^{-\sin(x)/2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{-\sin(x)/2}v) = -\sin(x)\cos(x)e^{-\sin(x)/2}$$

$$\begin{aligned}
e^{-\sin(x)/2}v &= \int -\sin(x)\cos(x)e^{-\sin(x)/2}\,\mathrm{d}x, \quad u = \frac{\sin(x)}{2} \\
&= -4\int ue^{-u}\,\mathrm{d}u \\
&= -4(-ue^{-u} - \int -e^{-u}\,\mathrm{d}u) \\
&= -4(-ue^{-u} - e^{-u} + C) \\
&= 2\sin(x)e^{-\sin(x)/2} + 4e^{-\sin(x)/2} + C
\end{aligned}$$

$$v = 2\sin(x) + 4 + Ce^{\sin(x)/2}$$

$$y = v^{-2} = (2\sin(x) + 4 + Ce^{\sin(x)/2})^{-2}$$

□

XII Slope being function of linear function

ODE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(u), \quad u = ax + by + c.$$

Solutions:

$$\int \frac{1}{a + bf(u)}\,\mathrm{d}u = x + C.$$

Proof.

$$\frac{\mathrm{d}u}{\mathrm{d}x} = a + b\frac{\mathrm{d}y}{\mathrm{d}x} = a + bf(u)$$

This is separable.

$$\int \frac{1}{a + bf(u)}\,\mathrm{d}u = x + C$$

□

XIII Slope being function of linear function over linear function

ODE:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = F(\frac{ax + by + c}{px + qy + r}), \quad aq - bp \neq 0.$$

We can find a unique solution of $ax + by + c = px + qy + r = 0$ for x and y, let it be $x = \alpha$, $y = \beta$, let

$$f(t) = F(\frac{a + bt}{p + qt}),$$

and we can rewrite it as:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(u), \quad u = \frac{y - \beta}{x - \alpha}.$$

Proof.

$$y = u(x - \alpha) + \beta$$

$$ax + by + c = (a + bu)(x - \alpha) + a\alpha + b\beta + c = (a + bu)(x - \alpha)$$

$$px + qy + r = (p + qu)(x - \alpha) + p\alpha + q\beta + r = (p + qu)(x - \alpha)$$

$$\frac{ax + by + c}{px + qy + r} = \frac{a + bu}{p + qu}$$

□

Solutions:

$$\int \frac{1}{f(u) - u}\,\mathrm{d}u = \ln|x - \alpha| + C,$$

and possibe constant or singular solutions:

$$f(u) = u.$$

Proof.

$$y = (x - \alpha)u + \beta$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\partial}{\partial x}((x - \alpha)u + \beta) + \frac{\partial}{\partial u}((x - \alpha)u + \beta)\frac{\mathrm{d}u}{\mathrm{d}x} = u + (x - \alpha)\frac{\mathrm{d}u}{\mathrm{d}x}$$

$$u + (x - \alpha)\frac{\mathrm{d}u}{\mathrm{d}x} = f(u)$$

$$\frac{1}{f(u) - u}\,\mathrm{d}u = \frac{1}{x - \alpha}\,\mathrm{d}x$$

This is separable.

$$\int \frac{1}{f(u) - u}\,\mathrm{d}u = \ln|x - \alpha| + C$$

□

3 Linear ODE

I Basics

i Linear ODE

An nth-order linear ODE of a unknown function or dependent variable $y : \mathbb{R} \to \mathbb{R}$ with respect to an independent variable $x \in \mathbb{R}$ can be written in the form, called normal form,

$$\sum_{i=0}^{n} a_i(x) y^{(i)} = f(x),$$

in which $a_i(x) : \Omega_i \subseteq \mathbb{R} \to \mathbb{R}$ are called coefficients and $f(x) : \Omega \subseteq \mathbb{R} \to \mathbb{R}^m$ is called forcing term.

ii Linear differential operator

A linear differential operator is a polynomial of the differential operators D, i.e., $\frac{\mathrm{d}}{\mathrm{d}x}$, and its order is the degree of the polynomial.

iii Linearity

Let L be the linear differential operator $L[\mathbf{y}] = \sum_{i=0}^{n} a_i(x)\mathbf{y}^{(i)}$. Then $L[\alpha\mathbf{y}_1 + \beta\mathbf{y}_2] = \alpha L[\mathbf{y}_1] + \beta L[\mathbf{y}_2]$. That's why it's called to be linear.

iv Homogeneity

A linear ODE is called a homogeneous linear ODE iff the forcing term is 0 for all x; otherwise, it is called a non-homogeneous linear ODE.

v Complementary or associated (homogeneous) equation, natural, complementary, or homogeneous solution or function, and particularly solution

Given an nth-order linear ODE

$$\sum_{i=0}^{n} a_i(x) y^{(i)} = f(x),$$

$$\sum_{i=0}^{n} a_i(x) y^{(i)} = 0,$$

is called its complementary or associated (homogeneous) equation, and the general solutions of the original equation is the sum of the general solutions y_c of the complementary equation, called natural, complementary, or homogeneous solution or function, and a particular solution y_p.

vi Superposition principle

If $y_1(x)$ and $y_2(x)$ are both solutions to a linear ODE on an interval, then $c_1 y_1(x) + c_2 y_2(x)$ is also a solution to it on the interval for any $c_1, c_2 \in \mathbb{R}$.

vii Wrońskian or Wronskian

For n functions $f_i : \Omega_i \subseteq \mathbb{R} \to \mathbb{R}$ for all $i \in \mathbb{N}_{\leq n}$ that are $(n-1)$ times differentiable on an interval I, the Wrońskian (also called Wronskian) of them, $W(f_1, f_2, \dots, f_n)$, is a function on I defined by

$$W(f_1, f_2, \dots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}.$$

viii Non-zero Wrońskian implies linear independence

For n functions $f_i : \Omega_i \subseteq \mathbb{R} \to \mathbb{R}$ for all $i \in \mathbb{N}_{\leq n}$ that are $(n-1)$ times differentiable on an interval I, if $W(f_1, f_2, \dots, f_n) \neq 0$, then they are linearly independent.

Proof. If they are not linearly independent, there exists not all zero $c_1, \dots, c_n$ such that

$$\sum_{i=1}^{n} c_i f_i = 0$$

$$\begin{bmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

All $c_i = 0$ if $W(f_1, f_2, \dots, f_n)(x) \neq 0$ by Cramer rule, thus $W(f_1, f_2, \dots, f_n)(x) = 0$. $\square$

ix General Abel's (differential equation) identity/formula

For n $(n-1)$ times differentiable solutions $y_1, y_2, \dots, y_n$ of $y^{(n)} + \sum_{i=0}^{n-1} p_i(x) y^{(i)} = 0$ on an interval I where $p_i(x)$ are continuous for all $i \in [0, n-1] \cap \mathbb{Z}$, for any $x_0 \in I$,

$$W(y_1, y_2, \dots, y_n)(x) = W(y_1, y_2, \dots, y_n)(x_0) \exp\left(-\int_{x_0}^{X} p_{n-1}(t) \, \mathrm{d}t \right).$$

Proof. Let W denote $W(y_1, y_2, \dots, y_n)$.

$$w(x) = \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix}.$$

$$
\begin{aligned}
W'(x) &= \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n)}(x) & y_2^{(n)}(x) & \cdots & y_n^{(n)}(x) \end{vmatrix} \\
&= \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n)}(x) + \sum_{i=0}^{n-2} p_i(x) {y_1}^{(i)} & y_2^{(n)}(x) + \sum_{i=0}^{n-2} p_i(x) {y_2}^{(i)} & \cdots & y_n^{(n)}(x) + \sum_{i=0}^{n-2} p_i(x) {y_n}^{(i)} \end{vmatrix} \\
&= \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ -p_{n-1}(x) y_1^{(n-1)}(x) & -p_{n-1}(x) y_2^{(n-1)}(x) & \cdots & -p_{n-1}(x) y_n^{(n-1)}(x) \end{vmatrix} \\
&= -p_{n-1}(x) W(x)
\end{aligned}
$$

Solve $W'(x) + p_{n-1}(x) W(x)$ using integrating factor $\exp\left(-\int_{x_0}^{x} p_{n-1}(t)\,\mathrm{d}t\right)$.

$$
W(x) = W(x_0) \exp\left(-\int_{x_0}^{X} p_{n-1}(t)\,\mathrm{d}t\right).
$$

□

x Non-zero Wrońskian of solutions of homogeneous linear equation iff linear independence of them

For n $(n-1)$ times differentiable solutions $y_1, y_2, \dots, y_n$ of $y^{(n)} + \sum_{i=0}^{n-1} p_i(x) y^{(i)} = 0$ on an interval I where $p_i(x)$ are continuous for all $i \in [0, n-1] \cap \mathbb{Z}$, for any $x_0 \in I$, $W(y_1, y_2, \dots, y_n)(x_0) \neq 0$ iff $y_1, y_2, \dots, y_n$ are linearly independent.

Proof. Abel's identity shows that

$$
W(y_1, y_2, \dots, y_n)(x) = W(y_1, y_2, \dots, y_n)(x_0) \exp\left(-\int_{x_0}^{X} p_{n-1}(t)\,\mathrm{d}t\right).
$$

$\exp\left(-\int_{x_0}^{X} p_{n-1}(t)\,\mathrm{d}t\right)$ is always non-zero. Thus $W(y_1, y_2, \dots, y_n)(x)$ is either zero for all $x \in I$ or non-zero for all $x \in I$.

If $y_1, y_2, \dots, y_n$ are linearly dependent, there exists $\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ such that

$$
\sum_{i=1}^{n} c_i y_i(x) = 0
$$

For any fixed $x_0 \in I$, let

$$\Omega = \begin{bmatrix} y_1(x_0) & y_2(x_0) & \cdots & y_n(x_0) \\ y_1'(x_0) & y_2'(x_0) & \cdots & y_n'(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x_0) & y_2^{(n-1)}(x_0) & \cdots & y_n^{(n-1)}(x_0) \end{bmatrix}$$

Differentiating $n-1$ times gives

$$\Omega \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Prove by contradiction. If $\det(\Omega) = 0$, by Cramer formula, $\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. □

xi Solutions of linear ODE as vector space

The general solutions to an nth-order linear ODE is a vector space of dimension n. A basis of it is called a fundamental set of solutions.

If $y_1(x), y_2(x), \ldots y_n(x)$ are n linearly independent solutions of an nth-order homogeneous linear ODE on an interval, then they are a basis of the vector space of the general solutions of the ODE on the interval, and $\sum_{i=1}^{n} c_i y_i(x)$ is the family of general solutions to it on the interval with n parameter $c_1, c_2, \ldots c_n$.

xii Solutions of non-homogeneous linear ODE as affine space

The general solutions to a non-homogeneous linear ODE is an affine space whose associated vector space is the vector space of the solutions to its complementary equation.

xiii Annihilator

An annihilator of a function f is a linear differential operator L such that $L(f) = 0$.

xiv Annihilator multiplication property

If P is an annihilator of f, then for any linear differential operator Q, PQ is an annihilator of f.

II Characteristic equation or auxiliary equation for constant coefficient homogeneous linear ODE

i Method of solution

Consider an nth-order constant coefficient homogeneous linear ODE

$$\sum_{j=0}^{n} a_j y^{(j)} = 0, \quad a_n \neq 0.$$

The formation of its general solutions using the characteristic equation (also called auxiliary equation) is as follows:

1. The characteristic equation is
$$\sum_{j=0}^{n} a_j r^j = 0.$$
2. Suppose there are k distinct real roots $r_1, r_2, \dots, r_k$, each r_j with multiplicity m_j, and h distinct pairs of conjugate imaginary roots $\alpha_1 \pm \beta_1 i, \alpha_2 \pm \beta_2 i, \dots, \alpha_h \pm \beta_h i$ ($\beta_j \neq 0$), each $\alpha_j \pm \beta_j i$ with multiplicity ω_j. (Thus, $\sum_{j=1}^{k} m_j + 2\sum_{j=1}^{h} \omega_j = n$.)
3. The n-parameter family of general solutions of the ODE is
$$y = \sum_{j=1}^{k} \sum_{u=0}^{m_j-1} A_{ju} x^u e^{r_j x} + \sum_{j=1}^{h} \sum_{u=0}^{\omega_j-1} (B_{ju} \cos(\beta_j x) + C_{ju} \sin(\beta_j x)) x^u e^{\alpha_j x},$$
with parameters A_{ju}, B_{ju}, C_{ju}.

ii Exponential order

For any solution $f(t)$ of any non-homogeneous linear ODE of one variable t, there exists some $M, T \geq 0$ and $\alpha \in \mathbb{R}$ such that for all $t > T$ $|f(t)| \leq e^{M\alpha t}$, and the smallest such α is the largest real part among all roots of its characteristic equation.

iii Example: simple harmonic motion

ODE:

$$\ddot{x} = -\omega^2 x, \quad \omega \in \mathbb{R}_{>0}.$$

Solutions:

$$x = A \sin(\omega t + \phi), \quad A, \phi \in \mathbb{R}.$$

Proof.

$$\ddot{x} + \omega^2 x = 0$$
$$r^2 + \omega^2 = 0$$
$$r = \pm \omega i$$
$$x(t) = A \sin(\omega t) + B \cos(\omega t) = A \sin(\omega t + \phi)$$

□

iv Example: linearly damped simple harmonic motion

ODE:

$$\ddot{x} = -k(x - C) - b\dot{x}, \quad k, b \in \mathbb{R}_{>0} \wedge C \in \mathbb{R}.$$

Solutions:

- Case $b^2 < 4k$ called underdamped:

$$x = Ae^{\frac{-b}{2}t} \sin\left(\frac{\sqrt{4k - b^2}}{2}t + \phi\right) + C.$$

- Case $b^2 = 4k$ called critically damped:

$$x = A(1 + Bt)e^{\frac{-b}{2}t} + C$$

- Case $b^2 > 4k$ called overdamped:

$$x = Ae^{\frac{-b}{2}t}(e^{\frac{\sqrt{b^2-4k}}{2}t} + Be^{\frac{-\sqrt{b^2-4k}}{2}t}) + C$$

Proof.

$$\lambda^2 + b\lambda + k = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4k}}{2}$$

Case $b^2 < 4k$:

$$\lambda = \frac{-b \pm \sqrt{4k - b^2}i}{2}$$

$$x - C = D(e^{\frac{-b+\sqrt{4k-b^2}i}{2}t} + Ee^{\frac{-b-\sqrt{4k-b^2}i}{2}t}) = Ae^{\frac{-b}{2}t} \sin\left(\frac{\sqrt{4k - b^2}}{2}t + \phi\right)$$

Case $b^2 = 4k$:

$$\lambda = \frac{-b}{2}$$

$$x - C = A(1 + Bt)e^{\frac{-b}{2}t}$$

Case $b^2 > 4k$:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4k}}{2}$$

$$x - C = A(e^{\frac{-b+\sqrt{b^2-4k}}{2}t} + Be^{\frac{-b-\sqrt{b^2-4k}}{2}t}) = Ae^{\frac{-b}{2}t}(e^{\frac{\sqrt{b^2-4k}}{2}t} + Be^{\frac{-\sqrt{b^2-4k}}{2}t})$$

□

III Method of undetermined coefficients (MUC) for constant coefficient non-homogeneous linear ODE

i Superposition approach

Consider an nth-order constant coefficient non-homogeneous linear ODE

$$\sum_{j=0}^{n} a_j y^{(j)} = g(x), \quad a_n \neq 0,$$

with $g(x)$ being able to be written in the form

$$g(x) = \sum_{j=1}^{p} k_j x^{m_j} e^{\alpha_j x} + \sum_{j=p+1}^{p+q} x^{m_j} e^{\alpha_j x}(k_j \cos(\beta_j x) + l_j \sin(\beta_j x)),$$

for some constants $p, q, m_j \in \mathbb{N}_0$ and $\alpha_j, \beta_j, k_j, l_j \in \mathbb{R}$.

The formation of its general solutions using the method of undetermined coefficients is as follows:

1. Solve the complementary equation

   $$\sum_{j=0}^{n} a_j y^{(j)} = 0, \quad a_n \neq 0$$

   to obtain its general solutions y_c.
2. For each term $k_j x^{m_j} e^{\alpha_j x}$ of $g(x)$ for $j \in \mathbb{N} \wedge j \leq p$, construct a particular solution y_j:
   (a) If α_j is not a root of the characteristic equation of the complementary equation, let $s_j = 0$; otherwise, let s_j be the multiplicity of α_j as a root of the characteristic equation of the complementary equation.

   (b) The ansatz is

   $$y_j = x^{s_j} e^{\alpha_j x} \sum_{k=0}^{m_j} C_{jk} x^k,$$

   with constants $C_{jk} \in \mathbb{R}$.

   (c) Solve for C_{jk} such that y_j is a particular solution of the constant coefficient non-homogeneous linear ODE

   $$\sum_{j=0}^{n} a_j y^{(j)} = k_j x^{m_j} e^{\alpha_j x}.$$

3. For each term $x^{m_j} e^{\alpha_j x}(k_j \cos(\beta_j x) + l_j \sin(\beta_j x))$ of $g(x)$ for $j \in \mathbb{N} \wedge p < j \leq p + q$, construct a particular solution y_j:

   (a) If $\alpha_j \pm \beta_j i$ is not a pair of conjugate imaginary roots of the characteristic equation of the complementary equation, let $s_j = 0$; otherwise, let s_j be the multiplicity of $\alpha_j \pm \beta_j i$ as a pair of conjugate imaginary roots of the characteristic equation of the complementary equation.

(b) The ansatz is

$$y_j = x^{s_j} e^{\alpha_j x} \sum_{k=0}^{m_j} (C_{jk} \cos(\beta_j x) + D_{jk} \sin(\beta_j x)) x^k,$$

with constants $C_{jk}, D_{jk} \in \mathbb{R}$.

(c) Solve for C_{jk}, D_{jk} such that y_j is a particular solution of the constant coefficient non-homogeneous linear ODE

$$\sum_{j=0}^{n} a_j y^{(j)} = x^{m_j} e^{\alpha_j x} (k_j \cos(\beta_j x) + l_j \sin(\beta_j x)).$$

4. A particular solution of the origin ODE is

$$y_p = \sum_{j=1}^{p+q} y_j,$$

and the general solutions of the origin ODE is

$$y = y_c + y_p.$$

ii Annihilator approach

Consider an nth-order constant coefficient non-homogeneous linear ODE

$$\sum_{j=0}^{n} a_j y^{(j)} = g(x), \quad a_n \neq 0,$$

with $g(x)$ being able to be written in the form

$$g(x) = \sum_{j=1}^{p} k_j x^{m_j} e^{\alpha_j x} + \sum_{j=p+1}^{p+q} x^{m_j} e^{\alpha_j x} (k_j \cos(\beta_j x) + l_j \sin(\beta_j x)),$$

for some constants $p, q, m_j \in \mathbb{N}_0$ and $\alpha_j, \beta_j, k_j, l_j \in \mathbb{R}$.

The formation of its general solutions using the method of undetermined coefficients is as follows:

1. Solve the complementary equation

$$\sum_{j=0}^{n} a_j y^{(j)} = 0, \quad a_n \neq 0$$

to obtain its general solutions y_c.

2. For each term $k_j x^{m_j} e^{\alpha_j x}$ of $g(x)$, a lowest-order annihilator is $(D - \alpha_j)^{m_j+1}$

Proof. Prove by Laplace transform.

$$\mathcal{L}\{k_j x^{m_j} e^{\alpha_j x}\} = \frac{k_j m_j!}{(s - \alpha_j)^{m_j+1}}$$

□

3. For each term $x^{m_j}e^{\alpha_j x}(k_j \cos(\beta_j x) + l_j \sin(\beta_j x))$ of $g(x)$, a lowest-order annihilator is $(D^2 - 2\alpha_j D + {\alpha_j}^2 + {\beta_j}^2)^{m_j+1}$

   Proof. Prove by Laplace transform.

$$\begin{aligned}\mathcal{L}\{x^{m_j}e^{\alpha_j x}(k_j \cos(\beta_j x) + l_j \sin(\beta_j x))\} &= \mathcal{L}\{x^{m_j}e^{\alpha_j x}((k_j + l_j)e^{i\beta_j x} - (k_j - l_j)ie^{-i\beta_j x})\} \\ &= (k_j + l_j)\mathcal{L}\{x^{m_j}e^{\alpha_j x + i\beta_j x}\} - (k_j - l_j)i\mathcal{L}\{x^{m_j}e^{\alpha_j x - i\beta_j x}\} \\ &= \frac{(k_j + l_j)m_j!}{(s - \alpha_j - i\beta_j)^{m_j+1}} - \frac{(k_j - l_j)im_j!}{(s - \alpha_j + i\beta_j)^{m_j+1}}\end{aligned}$$

   □

4. Find a lowest-order annihilator of $g(x)$, L_1, and apply it on both sides to obtain a homogeneous linear ODE

$$L_1(\sum_{j=0}^{n} a_j y^{(j)}) = 0.$$

5. Solve it and obtain its general solutions.
6. Substitute the solutions into the original ODE to obtain an explicit particular solution y_p, and the general solutions of the origin ODE is

$$y = y_c + y_p.$$

iii Example: Forced oscillation and resonance

TODO example with both approaches TODO

IV Variation of parameters or variation of constants method of solution of non-homogeneous linear equation

i Method of solution

Given an nth-order non-homogeneous linear ODE

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = q(x),$$

where $p_i(x)$ for all $i \in [0, n-1] \cap \mathbb{Z}$ are continuous on An interval I.

1. Let $y_1(x), y_2(x), \dots, y_n(x)$ be a basis of the vector space of the solutions of the complementary equation,

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = 0.$$

2. The particular solution y_p is given by:

$$y_p = \sum_{i=1}^{n} y_i(x) \int_{x_0}^{x} \frac{W_i(t)}{W(t)}\, \mathrm{d}t\,,$$

   where

- x_0 is an arbitrary point in I,
- $W(x)$ is the Wrońskian of $y_1(x), y_2(x), \dots, y_n(x)$,
- $W_i(x)$ is the Wrońskian of $W(x)$ but with the ith column replaced by $(0, 0, \dots, 0, q(x))$.

3. The general solutions are given by

$$y = y_c + y_p.$$

Proof. The particular solution is given by

$$y_p = \sum_{i=1}^{n} c_i(x) y_i(x),$$

where $c_i(x)$ are differentiable functions which are assumed to satisfy the $n-1$ conditions

$$\sum_{i=1}^{n} c_i'(x) y_i^{(j)}(x) = 0$$

for all $j \in \mathbb{N}_0 \wedge j \leq n-2$.

By repeated differentiation, we obtain:

$$y_p^{(j)} = \sum_{i=1}^{n} c_i(x) y_i^{(j)}(x)$$

for all $j \in \mathbb{N}_0 \wedge j \leq n-1$.

One last differentiation gives

$$y_p^{(n)} = \sum_{i=1}^{n} c_i(x) y_i^{(n)}(x) + \sum_{i=1}^{n} c_i'(x) y_i^{(n-1)}(x).$$

Substituting y_p into the original ODE gives:

$$y_p^{(n)} + \sum_{i=0}^{n-1} p_i(x) y_p^{(i)} = q(x),$$

$$\sum_{i=1}^{n} c_i(x) y_i^{(n)}(x) + \sum_{i=1}^{n} c_i'(x) y_i^{(n-1)}(x) + \sum_{i=0}^{n-1} p_i(x) \sum_{i=1}^{n} c_i(x) y_i^{(j)}(x) = q(x),$$

$$\sum_{i=1}^{n} c_i'(x) y_i^{(n-1)}(x) + \sum_{i=1}^{n} c_i(x)(y_i^{(n)}(x) + \sum_{i=0}^{n-1} p_i(x) y_i^{(j)}(x)) = q(x),$$

$$\sum_{i=1}^{n} c_i'(x) y_i^{(n-1)}(x) = q(x).$$

We obtain a system of linear equations

$$\sum_{i=1}^{n} c_i'(x) y_i^{(j)}(x) = 0$$

for all $j \in \mathbb{N}_0 \wedge j \leq n-2$, and

$$\sum_{i=1}^{n} c_i'(x) y_i^{(n-1)}(x) = q(x).$$

By Cramer Rule, we have

$$c_i'(x) = \frac{W_i(x)}{W(x)}, \quad i \in \mathbb{N} \wedge i \leq n.$$

□

ii For second-order equation, Green's function

For second-order equation

$$y'' + p_1(x)y' + p_0(x)y = q(x).$$

Let $y_1(x), y_2(x)$ be two linearly independent solutions of its complementary equation

$$y'' + p_1(x)y' + p_0(x)y = 0.$$

The particular solution (or family of general solutions with the integral constants being parameters) is given by

$$y_p(x) = -y_1(x) \int_{x_0}^{x} \frac{y_2(t)q(t)}{y_1(t)y_2'(t) - y_2(t)y_1'(t)} \,\mathrm{d}t + y_2(x) \int_{x_0}^{x} \frac{y_1(t)q(t)}{y_1(t)y_2'(t) - y_2(t)y_1'(t)} \,\mathrm{d}t\,.$$

This can be written as

$$y_p(x) = \int_{x_0}^{x} G(x,t)q(t) \,\mathrm{d}t\,,$$

where

$$G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{y_1(t)y_2'(t) - y_2(t)y_1'(t)}$$

and is called the Green's function for the ODE and the linear differential operator $D^2 + p_1(x)D + p_0(x)$.

V Reduction of order

i Method

Given an nth-order homogeneous linear ODE

$$\sum_{i=0}^{n} p_i(x) y^{(i)} = 0,$$

if a nontrivial particular solution $y_1(x)$ has been known, we can reduce the order of the ODE by 1 by the substitution

$$y = vy_1$$

and then the substitution

$$u = v'$$

to an $(n-1)$th-order linear ODE.

ii Closed-form formula for second-order homogeneous linear ODEs

Given a second-order homogeneous linear ODE

$$y'' + p(x)y' + q(x)y = 0,$$

if a nontrivial particular solution $y_1(x)$ that has been known, then the general solutions of the ODE is

$$y = y_1 \int \frac{e^{-\int p(x)\mathrm{d}x}}{y_1(x)^2}\,\mathrm{d}x\,.$$

Proof. Substitute

$$\begin{aligned} y &= vy_1, \\ y' &= v'y_1 + vy_1', \\ y'' &= v''y_1 + 2v'y_1' + vy_1'', \\ y_1'' &= -py_1' - qy_1, \\ y'' &= -py' - qy =, \\ -pv'y_1 - pvy_1' - qvy_1 &= v''y_1 + 2v'y_1' - pvy_1' - qvy_1, \\ y_1v'' + (2y_1' + py_1)v' &= 0, \end{aligned}$$

Substitute

$$\begin{aligned} u &= v', \\ y_1u' + (2y_1' + py_1)u &= 0, \\ u' + (\frac{2y_1'}{y_1} + p)u &= 0, \\ \mu(x) = \exp\left(\int \frac{y_1'}{y_1} + p\,\mathrm{d}x\right) &= y_1(x)^2 e^{\int p(x)\mathrm{d}x}, \\ (u\mu(x))' &= 0, \\ u = \frac{C}{\mu(x)} = C\frac{e^{-\int p(x)\mathrm{d}x}}{y_1(x)^2} &= \frac{e^{-\int p(x)\mathrm{d}x}}{y_1(x)^2}, \\ v = \int u\,\mathrm{d}x &= \int \frac{e^{-\int p(x)\mathrm{d}x}}{y_1(x)^2}\,\mathrm{d}x\,, \\ y = vy_1 &= y_1 \int \frac{e^{-\int p(x)\mathrm{d}x}}{y_1(x)^2}\,\mathrm{d}x\,. \end{aligned}$$

□

iii Example:

TODO

VI (Cauchy-)Euler or equidimensional equation

i Definition

An nth order Cauchy-Euler equation has the form

$$\sum_{i=0}^{n} a_i x^i y^{(i)} = g(x), \quad a_n \neq 0.$$

ii Method of solution

Consider an nth-order Cauchy-Euler equation

$$\sum_{i=0}^{n} a_i x^i y^{(i)} = g(x), \quad a_n \neq 0.$$

The formation of its general solutions using the characteristic equation (also called auxiliary equation) is as follows:

1. The complementary equation is
$$\sum_{i=0}^{n} a_i x^i y^{(i)} = 0$$
2. The characteristic equation is
$$\sum_{j=0}^{n} a_j r^j = 0.$$
3. Suppose there are k distinct real roots $r_1, r_2, \ldots, r_k$, each r_j with multiplicity m_j, and h distinct pairs of conjugate imaginary roots $\alpha_1 \pm \beta_1 i, \alpha_2 \pm \beta_2 i, \ldots, \alpha_h \pm \beta_h i$ ($\beta_j \neq 0$), each $\alpha_j \pm \beta_j i$ with multiplicity ω_j. (Thus, $\sum_{j=1}^{k} m_j + 2\sum_{j=1}^{h} \omega_j = n$.)
4. The n-parameter family of solutions of the complementary equation is
$$y_c = \sum_{j=1}^{k} \sum_{u=0}^{m_j-1} A_{ju} (\ln x)^u x^{r_j} + \sum_{j=1}^{h} \sum_{u=0}^{\omega_j-1} (B_{ju} \cos\left(\beta_j \ln x\right) + C_{ju} \sin\left(\beta_j \ln x\right))(\ln x)^u x^{\alpha_j},$$
with parameters A_{ju}, B_{ju}, C_{ju}.
5. Then, we use variation of parameters. Let $y_1(x), y_2(x), \ldots, y_n(x)$ be a basis of the vector space of y_c.

   The particular solution y_p is given by:
$$y_p = \sum_{i=1}^{n} y_i(x) \int_{x_0}^{x} \frac{W_i(t)}{W(t)} \, \mathrm{d}t\,,$$
   where

   - x_0 is an arbitraty point in the interval of interest,

- $W(x)$ is the Wrońskian of $y_1(x), y_2(x), \ldots, y_n(x)$,
- $W_i(x)$ is the Wrońskian of $W(x)$ but with the ith column replaced by $(0, 0, \ldots, 0, g(x))$.

6. The general solutions are given by
$$y = y_c + y_p.$$

VII Power series solutions and Frobenius method for second-order homogeneous linear ODE

i Type of Points

Consider a second-order homogeneous linear ODE

$$f(x)y'' + g(x)y' + h(x)y = 0$$

where f, g, h are continuous on interval I.

Let
$$p(x) = \frac{g(x)}{f(x)}, \quad q(x) = \frac{h(x)}{f(x)}.$$

For any $a \in I$, it is called an ordinary point iff $p(x)$ and $q(x)$ are both analytic at a and called a singular point otherwise.

For a singular point a, it is called a regular singular point iff $(x-a)p(x)$ and $(x-a)^2 q(x)$ are both analytic at a and called an irregular singular point otherwise.

ii Power series solution or solution about ordinary points

Consider a second-order homogeneous linear ODE

$$f(x)y'' + g(x)y' + h(x)y = 0$$

where f, g, h are continuous on interval I.

Let
$$p(x) = \frac{g(x)}{f(x)}, \quad q(x) = \frac{h(x)}{f(x)}.$$

1. We seek a power series solution of
$$y'' + p(x)y' + q(x)y = 0$$
in the form:
$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n,$$
where x_0 is a fixed ordinary point called expansion point and a_n are constants to be determined. There are always two linearly independent solutions of this form.

2. Differentiate the series term by term:
$$y' = \sum_{n=0}^{\infty} a_n n (x - x_0)^{n-1} = \sum_{n=0}^{\infty} a_{n+1}(n+1)(x - x_0)^n,$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1)(x-x_0)^{n-2} = \sum_{n=0}^{\infty} a_{n+2}(n+2)(n+1)(x-x_0)^n.$$

3. Substitute into the ODE, rewrite $p(x)$ and $q(x)$ into Taylor expansion at x_0, rewrite the infinite series products (if any) into Cauchy products, and rewrite the LHS as a power series in x^n.
4. Solve for the recurrence relation of a_n. We may set a_0 as any nonzero constant, typically 1.
5. Let R be the distance between x_0 and the nearest singular point if there is one or ∞ if there is none. The two-parameter family of general solutions for all $0 < |x - x_0| < R$ with parameters a_0, a_1 is:
$$y = \sum_{n=0}^{\infty} a_n (x-x_0)^n.$$

Proof. TODO □

iii Frobenius method or solution about singular points

Consider a second-order homogeneous linear ODE

$$f(x)y'' + g(x)y' + h(x)y = 0.$$

Let
$$p(x) = \frac{g(x)}{f(x)}, \quad q(x) = \frac{h(x)}{f(x)}.$$

1. We seek a solution, called Frobenius series, of
$$(x-x_0)^2 y'' + (x-x_0)r(x)y' + s(x)y = 0$$
in the form:
$$y = \sum_{n=0}^{\infty} a_n (x-x_0)^{n+r}, \quad a_0 \neq 0,$$
where x_0 is a fixed regular singular point called expansion point, $r(x) = (x-x_0)p(x)$ and $s(x) = (x-x_0)^2 q(x)$ are analytic at x_0, and a_n are constants to be determined.
2. Differentiate the series term by term:
$$y' = \sum_{n=0}^{\infty} a_n (n+r)(x-x_0)^{n+r-1},$$
$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1)(x-x_0)^{n+r-2}.$$
3. Substitute into the ODE, rewrite $(x-x_0)r(x)$ and $s(x)$ into Taylor expansion at x_0, rewrite the infinite series products (if any) into Cauchy products, and let the sum of all terms involving a_0 of the three be L.

4. The indicial polynomial $I(r)$ is defined as the coefficient of the lowest power term of $\frac{L}{a_0}$, and the indicial equation is defined as

$$I(r) = 0,$$

which is a quadratic equation for r.

5. Solve the indicial equation for r. Let the roots be $r_1 \geq r_2$, called indical roots or exponents.

6. Let R be the distance between x_0 and the nearest singular point if there is one or ∞ if there is none. Depending on the relation of r_1 and r_2:

   - Case $r_1 - r_2 \notin \mathbb{Z} \wedge r_1, r_2 \in \mathbb{R}$: Two linearly independent solutions for all $0 < |x - x_0| < R$ are:

$$y_1(x) = (x - x_0)^{r_1} \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

   - Case $r_1, r_2 = \alpha \pm \beta i, \quad \beta \neq 0$: Two linearly independent solutions for all $0 < |x - x_0| < R$ are:

$$y_1(x) = (x - x_0)^{\alpha} \cos\left(\beta \ln|x - x_0|\right) \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^{\alpha} \sin\left(\beta \ln|x - x_0|\right) \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

   - Case $r_1 - r_2 \in \mathbb{N}_0$: Two solutions for all $0 < |x - x_0| < R$ are:

$$y_1(x) = (x - x_0)^{r_1} \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

     If they are linearly independent, no modification is needed; otherwise, y_2 is modified to

$$y_2(x) = \ln|x - x_0| y_1(x) + (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R},$$

     where B_n is different from the original one.

7. Substitute into the ODE, rewrite $(x - x_0) r(x)$ and $s(x)$ into Taylor expansion at x_0, rewrite the infinite series products (if any) into Cauchy products, and rewrite the LHS as a power series in x^n.

8. Solve for the recurrence relations of A_n and B_n.

   - Case $(r_1 - r_2 \notin \mathbb{Z} \wedge r_1, r_2 \in \mathbb{R}) \vee (r_1, r_2 = \alpha \pm \beta i, \quad \beta \neq 0)$: We may set A_0 and B_0 as any nonzero constants (typically 1).

- Case $r_1 - r_2 \in \mathbb{N}_0$: The terms multiplied by $\ln\left|x - x_0\right|$ in the substitution of $y_2(x)$ yield the same recurrence relation for A_n as that obtained from the substitution of $y_1(x)$, and therefore need not be solved again. We may set A_0 as any nonzero constant (typically 1). The lowest two-order equations in the substitution of $y_2(x)$ may determine B_0 and/or B_1, or relation of B_0 and B_1. If one of them remains free after applying the constraint, it may be chosen arbitrarily with the constraint that B_0 can't be 0, and the remaining can be determined uniquely by the recurrence relation.

9. The two-parameter family of general solutions for all $0 < |x - x_0| < R$ is:

$$y = C_1 y_1(x) + C_2 y_2(x), \quad C_1, C_2 \in \mathbb{R}.$$

Proof. TODO □

iv Example: $y'' - 2xy' + y = 0$

ODE:

$$y'' - 2xy' + y = 0.$$

Solutions:

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

$$a_0, a_1 \in \mathbb{R}, \quad a_{n+2} = \frac{n-1}{(n+2)(n+1)} a_n.$$

Proof.

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=0}^{\infty} a_{n+1}(n+1)x^n$$

$$y'' = \sum_{n=0}^{\infty} a_{n+2}(n+2)(n+1)x^n$$

$$a_{n+2}(n+2)(n+1) - a_n n + a_n = 0$$

$$a_{n+2} = \frac{n-1}{(n+2)(n+1)} a_n$$

□

v Example: $y'' + \sin(x)y' + x^2 y = 0$

ODE:

$$y'' + \sin(x)y' + x^2 y = 0.$$

Solutions:

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

where:

$$a_{-1} = a_{-2} = 0, \quad a_0, a_1 \in \mathbb{R},$$

$$a_{n+2}(n+2)(n+1)+\sum_{k=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^k}{(2k+1)!}a_{n-2k}(n-2k)+a_{n-2}=0.$$

Proof.

$$y=\sum_{n=0}^{\infty}a_nx^n$$

$$y'=\sum_{n=0}^{\infty}a_{n+1}(n+1)x^n$$

$$y''=\sum_{n=0}^{\infty}a_{n+2}(n+2)(n+1)x^n$$

$$\sin(x)=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}x^{2n+1}$$

$$\sum_{n=0}^{\infty}a_{n+2}(n+2)(n+1)x^n+\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}x^{2n+1}\sum_{n=0}^{\infty}a_{n+1}(n+1)x^n+x^2\sum_{n=0}^{\infty}a_nx^n=0.$$

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}x^{2n+1}\sum_{n=0}^{\infty}a_{n+1}(n+1)x^n&=\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{(-1)^k}{(2k+1)!}x^{2k+1}a_{n-k+1}(n-k+1)x^{n-k}\\&=\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{(-1)^k}{(2k+1)!}a_{n-k+1}(n-k+1)x^{n+k+1}\\&=\sum_{n=0}^{\infty}\sum_{k=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^k}{(2k+1)!}a_{n-2k}(n-2k)x^n\end{aligned}$$

$$\sum_{n=0}^{\infty}(a_{n+2}(n+2)(n+1)+\sum_{k=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^k}{(2k+1)!}a_{n-2k}(n-2k)+a_{n-2})x^n=0,\quad a_{-1}=a_{-2}=0$$

$$a_{n+2}(n+2)(n+1)+\sum_{k=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^k}{(2k+1)!}a_{n-2k}(n-2k)+a_{n-2}=0$$

□

vi Example: $(x-x_0)^2y''+(x-x_0)y'-y=0$

ODE:

$$(x-x_0)^2y''+(x-x_0)y'-y=0.$$

Solutions:

$$y(x)=A(x-x_0)+B(x-x_0)^{-1},\quad A,B\in\mathbb{R}.$$

Proof.

$$y=\sum_{n=0}^{\infty}a_n(x-x_0)^{n+r}$$

$$y' = \sum_{n=0}^{\infty} a_n(n+r)(x-x_0)^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)(x-x_0)^{n+r-2}$$

$$(x-x_0)^2 y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)(x-x_0)^{n+r}$$

$$(x-x_0)y' = \sum_{n=0}^{\infty} a_n(n+r)(x-x_0)^{n+r}$$

$$L = a_0 r(r-1)(x-x_0)^r + a_0 r(x-x_0)^r - a_0(x-x_0)^r$$

$$I(r) = a_0 r(r-1) + a_0 r - a_0 = a_0(r^2-1) = 0$$

$$r = \pm 1$$

$$y_1(x) = \sum_{n=0}^{\infty} A_n(x-x_0)^{n+1}$$

$$y_1'(x) = \sum_{n=0}^{\infty} A_n(n+1)(x-x_0)^n$$

$$y_1''(x) = \sum_{n=0}^{\infty} A_n(n+1)n(x-x_0)^{n-1}$$

$$(x-x_0)y_1'(x) = \sum_{n=0}^{\infty} A_n(n+1)(x-x_0)^{n+1}$$

$$(x-x_0)^2 y_1''(x) = \sum_{n=0}^{\infty} A_n(n+1)n(x-x_0)^{n+1}$$

$$A_n(n+1)n + A_n(n+1) - A_n = 0.$$

$$A_n = 0, \quad \forall n \in \mathbb{N}$$

$$y_2(x) = \sum_{n=0}^{\infty} B_n(x-x_0)^{n-1}$$

$$y_2'(x) = \sum_{n=0}^{\infty} B_n(n-1)(x-x_0)^{n-2}$$

$$y_2''(x) = \sum_{n=0}^{\infty} B_n(n-1)(n-2)(x-x_0)^{n-3}$$

$$\sum_{n=0}^{\infty} B_n(n-1)(n-2)(x-x_0)^{n-1} + B_n(n-1)(x-x_0)^{n-1} - B_n(x-x_0)^{n-1} = 0$$

$$\sum_{n=0}^{\infty} B_n(n+1)(n-1)(x-x_0)^n = 0$$

$$B_n(n+1)(n-1) = 0$$

$$B_n = 0, \quad \forall n \in \mathbb{N} \wedge n \neq 2$$

$$y_2(x) = B_0(x-x_0)^{-1} + B_2(x-x_0)$$

$$y(x) = C_1 y_1(x) + C_2 y_2(x) = A(x-x_0) + B(x-x_0)^{-1}, \quad A, B \in \mathbb{R}$$

□

vii Example: $xy'' + (x-1)^2 y' + y = 0$

ODE:

$$xy'' + (x-1)^2 y' + y = 0.$$

Solutions:

$$y(x) = \sum_{n=0}^{\infty} (C_1 A_n + C_2 B_n) x^n + C_2 \ln|x| \sum_{n=0}^{\infty} A_n x^n,$$

$$A_0 = 1, \quad A_1 = -1, \quad B_0 = 4, \quad B_1 = 0, \quad C_1, C_2 \in \mathbb{R},$$

$$A_{n+2}(n+2)^2 - A_{n+1}(2n+1) + A_n n = 0,$$

$$2A_{n+2}(n+2) + B_{n+2}(n+2)^2 - 2A_{n+1} - B_{n+1}(2n+1) + A_n + B_n n = 0.$$

Proof.

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2}$$

$$xy'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-1}$$

$$(x-1)^2 y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r+1} - 2\sum_{n=0}^{\infty} a_n (n+r) x^{n+r} + \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$L = a_0 r(r-1) x^{r-1} + a_0 r x^{r+1} - 2a_0 r x^r + a_0 r x^{r-1}$$

$$I(r) = a_0 r(r-1) + a_0 r = 0$$

$$r_1 = r_2 = 0$$

$$y_1(x) = \sum_{n=0}^{\infty} A_n x^n$$

$$\sum_{n=0}^{\infty} A_n n(n-1) x^{n-1} + \sum_{n=0}^{\infty} A_n n x^{n+1} - 2\sum_{n=0}^{\infty} A_n n x^n + \sum_{n=0}^{\infty} A_n n x^{n-1} + \sum_{n=0}^{\infty} A_n x^n = 0$$

$$\sum_{n=0}^{\infty} A_{n+1}(n+1) n x^n + \sum_{n=0}^{\infty} A_{n-1}(n-1) x^n - 2\sum_{n=0}^{\infty} A_n n x^n + \sum_{n=0}^{\infty} A_{n+1}(n+1) x^n + \sum_{n=0}^{\infty} A_n x^n = 0$$

$$A_{n+1}(n+1)^2 - A_n(2n-1) + A_{n-1}(n-1) = 0$$

$$A_1 - A_0(-1) = 0, \quad A_1 = -A_0$$

$$A_{n+2}(n+2)^2 - A_{n+1}(2n+1) + A_n n = 0$$

$$y_2(x) = \ln|x| \sum_{n=0}^{\infty} A_n x^n + \sum_{n=0}^{\infty} B_n x^n$$

$$y_2'(x) = \ln|x| \sum_{n=0}^{\infty} A_n n x^{n-1} + \sum_{n=0}^{\infty} A_n x^{n-1} + \sum_{n=0}^{\infty} B_n n x^{n-1}$$

$$y_2''(x) = \ln|x| \sum_{n=0}^{\infty} A_n n(n-1) x^{n-2} + \sum_{n=0}^{\infty} A_n n x^{n-2} + \sum_{n=0}^{\infty} A_n (n-1) x^{n-2} + \sum_{n=0}^{\infty} B_n n(n-1) x^{n-2}$$

$$x y_2''(x) = \ln|x| \sum_{n=0}^{\infty} A_n n(n-1) x^{n-1} + \sum_{n=0}^{\infty} A_n (2n-1) x^{n-1} + \sum_{n=0}^{\infty} B_n n(n-1) x^{n-1}$$

$$x^2 y_2'(x) = \ln|x| \sum_{n=0}^{\infty} A_n n x^{n+1} + \sum_{n=0}^{\infty} A_n x^{n+1} + \sum_{n=0}^{\infty} B_n n x^{n+1}$$

$$x y_2'(x) = \ln|x| \sum_{n=0}^{\infty} A_n n x^n + \sum_{n=0}^{\infty} A_n x^n + \sum_{n=0}^{\infty} B_n n x^n$$

$$\sum_{n=0}^{\infty} A_n(2n-1)x^{n-1} + \sum_{n=0}^{\infty} B_n n(n-1)x^{n-1} + \sum_{n=0}^{\infty} A_n x^{n+1} + \sum_{n=0}^{\infty} B_n n x^{n+1} - 2\sum_{n=0}^{\infty} A_n x^n - 2\sum_{n=0}^{\infty} B_n n x^n + \sum_{n=0}^{\infty} A_n x^{n-1} + \sum_{n=0}^{\infty} B_n n x^{n-1}$$

$$\sum_{n=0}^{\infty} A_n 2n x^{n-1} + B_n n^2 x^{n-1} - 2A_n x^n - B_n(2n-1)x^n + A_n x^{n+1} + B_n n x^{n+1} = 0$$

$$-2A_0 + B_0 + 2A_1 + B_1 = 0, \quad B_0 + B_1 = 4A_0$$

$$2A_{n+2}(n+2) + B_{n+2}(n+2)^2 - 2A_{n+1} - B_{n+1}(2n+1) + A_n + B_n n = 0$$

Set

$$A_0 = 1, \quad A_1 = -1, \quad B_0 = 4, \quad B_1 = 0$$

$$y(x) = C_1 y_1(x) + C_2 y_2(x) = \sum_{n=0}^{\infty} (C_1 A_n + C_2 B_n) x^n + C_2 \ln|x| \sum_{n=0}^{\infty} A_n x^n$$

□

viii Example for complex roots: TODO

TODO

VIII Laplace transform

i Constant coefficients

To solve

$$\sum_{i=0}^{n} a_i y^{(i)} = g(x)$$

assuming that y is of exponential order.

Laplace transform both sides:

$$\sum_{j=0}^{n} a_j(s^j Y(s) - \sum_{i=0}^{j-1} s^{j-1-i}\frac{\mathrm{d}^i y}{\mathrm{d}t^i}(0)) = G(s)$$

Let

$$P(s) = \sum_{j=0}^{n} a_j s^j$$

$$Q(s) = \sum_{j=0}^{n} a_j \sum_{i=0}^{j-1} s^{j-1-i}\frac{\mathrm{d}^i y}{\mathrm{d}t^i}(0)$$

Thus

$$P(s)Y(s) = Q(s) + G(s),$$

where $P(s)$ and $Q(s)$ are polynomials of s of degrees n and $< n$.

We call

$$W(s) = \frac{1}{P(s)}$$

transfer function of the system and write

$$Y(s) = W(s)Q(s) + W(s)G(s)$$

$$y_0(t) = \mathcal{L}^{-1}\{W(s)Q(s)\}$$

$$y_1(t) = \mathcal{L}^{-1}\{W(s)G(s)\}$$

$$y(t) = y_0(t) + y_1(t)$$

We call $y_0(t)$ natural, homogeneous, unforced, or zero-input response (ZIR) and $y_1(t)$ forced, driven, particular, or zero-state responses (ZSR).

IX Bessel functions and Legendre polynomials

i Bessel functions

Bessel functions are a family of special functions that are solutions to the Bessel's equation:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0,$$

where $x \in \mathbb{R}$ is the independent variable and $\nu \in \mathbb{R}$ is called order.

- Bessel function of the first kind:

  $$J_\nu(x) = \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu}.$$

  - For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

    $$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+\nu+1)}(\frac{x}{2})^{2m+\nu}.$$

    $J_\nu(x)$ and $J_{-\nu}(x)$ are two linearly independent solutions of the Bessel's equation. $J_\nu(x)$ is defined on $\mathbb{R}_{>=0}$ for $\nu \in \mathbb{R}_{>0} \setminus \mathbb{Z}$, and on $\mathbb{R}_{>0}$ for $\nu \in \mathbb{R}_{<0} \setminus \mathbb{Z}$. $J_\nu(0) = 0$ for $\nu \in \mathbb{R}_{>0} \setminus \mathbb{Z}$.

- For $\nu \in \mathbb{Z}$:

$$J_\nu(x) = \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu},$$

  that is,

$$J_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m+|\nu|)!}(\frac{x}{2})^{2m+|\nu|}$$

$$J_{-|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m+|\nu|}}{m!(m+|\nu|)!}(\frac{x}{2})^{2m+|\nu|},$$

  that is,

  * For $\nu \in \mathbb{N}_0$:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m+\nu)!}(\frac{x}{2})^{2m+\nu}$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m+\nu}}{m!(m+\nu)!}(\frac{x}{2})^{2m+\nu}$$

  * For $-\nu \in \mathbb{N}_0$:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m-\nu}}{m!(m-\nu)!}(\frac{x}{2})^{2m-\nu}$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m-\nu)!}(\frac{x}{2})^{2m-\nu}$$

  and

$$J_{-\nu}(x) = J_\nu(-x) = (-1)^\nu J_\nu(x)$$

  $J_\nu(x)$ is defined on $\mathbb{R}$. $J_0(0) = 1$. $J_\nu(0) = 0$ for $\nu \in \mathbb{Z}_{\neq 0}$.

- Bessel function of the second kind:

$$Y_\nu(x) = \lim_{\mu\to\nu} \frac{J_\mu(x)\cos(\mu\pi) - J_{-\mu}(x)}{\sin(\mu\pi)}.$$

  - For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

$$Y_\nu(x) = \frac{J_\nu(x)\cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)},$$

    which is a linear combination of $J_\nu(x)$ and $J_{-\nu}(x)$.

  - For $\nu \in \mathbb{Z}$:

$$Y_\nu(x) = \lim_{\mu\to\nu} Y_\mu(x),$$

    which is a solution that is linearly independent to $J_\nu(x)$. $Y_\nu(x)$ is defined on $\mathbb{R}_{>0}$.

$$\lim_{x\to 0^+} Y_\nu(x) = -\infty.$$

$$Y_0(x) = \frac{2}{\pi}J_0(x)(\gamma + \ln\left(\frac{x}{2}\right)) - \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^k}{(k!)^2}(sum_{j=1}^{k}\frac{1}{j})(\frac{x}{2})^{2k}.$$

The general solution is:

- For $\nu \in \mathbb{R}$:

$$y(x) = CJ_\nu(x) + DY_\nu(x).$$

- For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

$$y(x) = CJ_\nu(x) + DJ_{-\nu}(x).$$

Proof.

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0$$

$$y'' + x^{-1}y' + (1 - \nu^2x^{-2})y = 0$$

0 is a regular singular point.

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} a_n(n+r)x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)x^{n+r-2}$$

$$\sum_{n=0}^{\infty} a_n(n+r)(n+r-1)x^{n+r} + \sum_{n=0}^{\infty} a_n(n+r)x^{n+r} + \sum_{n=0}^{\infty} a_n x^{n+r+2} - \nu^2 \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$L = a_0r(r-1)x^r + a_0rx^r + a_0x^{r+2} - \nu^2a_0x^r$$

$$I(r) = r^2 - \nu^2 = 0$$

$$r = \pm\nu$$

$$y_1(x) = \sum_{n=0}^{\infty} A_n x^{n+|\nu|}$$

$$y_1'(x) = \sum_{n=0}^{\infty} A_n(n+|\nu|)x^{n+|\nu|-1}$$

$$y_1''(x) = \sum_{n=0}^{\infty} A_n(n+|\nu|)(n+|\nu|-1)x^{n+|\nu|-2}$$

$$y_2(x) = \sum_{n=0}^{\infty} B_n x^{n-|\nu|}$$

$$y_2'(x) = \sum_{n=0}^{\infty} B_n(n-|\nu|)x^{n-|\nu|-1}$$

$$y_2''(x) = \sum_{n=0}^{\infty} B_n(n-|\nu|)(n-|\nu|-1)x^{n-|\nu|-2}$$

$$\sum_{n=0}^{\infty} A_n(n+|\nu|)(n+|\nu|-1)x^{n+|\nu|} + \sum_{n=0}^{\infty} A_n(n+|\nu|)x^{n+|\nu|} + \sum_{n=0}^{\infty} A_n x^{n+|\nu|+2} - |\nu|^2 \sum_{n=0}^{\infty} A_n x^{n+|\nu|} = 0$$

$$A_1(1 + 2|\nu|)x^{1+|\nu|} + \sum_{n=2}^{\infty}(A_n(n^2 + 2n|\nu|) + A_{n-2})x^{n+|\nu|} = 0$$

$$A_n = 0, \quad \frac{n+1}{2} \in \mathbb{N}$$

$$A_n(n^2 + 2n|\nu|) + A_{n-2} = 0, \quad \frac{n}{2} \in \mathbb{N}$$

$$A_{2m} = \frac{(-1)^m A_0}{\prod_{i=1}^{m}(2i)(2i+2|\nu|)} = \lim_{\mu \to |\nu|} \frac{(-1)^m A_0 \Gamma(\mu+1)}{2^{2m} m! \Gamma(m+\mu+1)}, \quad m \in \mathbb{N}$$

Let

$$A_0 = \lim_{\mu \to |\nu|} \frac{1}{2^\mu \Gamma(\mu+1)}$$

$$A_{2m} = \lim_{\mu \to |\nu|} \frac{(-1)^m}{2^{2m+\mu} m! \Gamma(m+\mu+1)}, \quad m \in \mathbb{N}$$

$$y_1(x) = J_{|\nu|}(x) = \sum_{m=0}^{\infty} \lim_{\mu \to |\nu|} \frac{(-1)^m}{2^{2m+\mu} m! \Gamma(m+\mu+1)} x^{2m+|\nu|} = \sum_{m=0}^{\infty} \lim_{\mu \to |\nu|} \frac{(-1)^m}{m! \Gamma(m+\mu+1)} (\frac{x}{2})^{2m+|\nu|}$$

$$y_2(x) = J_{-|\nu|}(x) = \sum_{m=0}^{\infty} \lim_{\mu \to |\nu|} \frac{(-1)^m}{m! \Gamma(m-\mu+1)} (\frac{x}{2})^{2m-|\nu|}$$

For $\nu \in \mathbb{R} \setminus \mathbb{Z}$, that is,

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\nu+1)} (\frac{x}{2})^{2m+\nu}$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m-\nu+1)} (\frac{x}{2})^{2m-\nu}$$

For $2\nu \in \mathbb{Z}$, test whether they are are linearly dependent.

$$J'_\nu(x) = \sum_{m=0}^{\infty} \lim_{\mu \to \nu} \frac{(-1)^m}{m! \Gamma(m+\mu+1)} (\frac{x}{2})^{2m+\nu-1} \frac{2m+\nu}{2}$$

$$J'_{-\nu}(x) = \sum_{m=0}^{\infty} \lim_{\mu \to \nu} \frac{(-1)^m}{m! \Gamma(m-\mu+1)} (\frac{x}{2})^{2m-\nu-1} \frac{2m-\nu}{2}$$

$$W(J_\nu, J_{-\nu})(x) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \lim_{\mu \to \nu} \frac{(-1)^{m+n}(m-n-\nu)}{m! n! \Gamma(m-\mu+1) \Gamma(n+\mu+1)} (\frac{x}{2})^{2m+2n-1}$$

The coefficient of the x^{-1} term is

$$\lim_{\mu \to \nu} \frac{-2\nu}{\Gamma(1-\mu)\Gamma(1+\mu)} = \lim_{\mu \to \nu} \frac{-2}{\Gamma(\mu)\Gamma(1-\mu)} = \frac{-2\sin(\pi\nu)}{\pi}$$

Thus for $\nu \in \mathbb{Z}$, $W(J_\nu, J_{-\nu}) = 0$,

$$J_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|}$$

and observe that for $k \in \mathbb{N}_0$,

$$\lim_{\mu \to \nu} \frac{1}{\Gamma(-k)} = 0,$$

thus

$$J_{-|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m+|\nu|}}{m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|},$$

and

$$J_{-|\nu|}(x) = (-1)^{|\nu|} J_{|\nu|}(x),$$

that is,

- For $\nu \in \mathbb{N}_0$:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m+\nu)!} (\frac{x}{2})^{2m+\nu}$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m+\nu}}{m!(m+\nu)!} (\frac{x}{2})^{2m+\nu}$$

$$J_{-\nu}(x) = (-1)^\nu J_\nu(x)$$

- For $-\nu \in \mathbb{N}_0$:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m-\nu}}{m!(m-\nu)!} (\frac{x}{2})^{2m-\nu}$$

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^{m}}{m!(m-\nu)!} (\frac{x}{2})^{2m-\nu}$$

$$J_{-\nu}(x) = (-1)^\nu J_\nu(x)$$

and a new $y_2(x)$ is

$$y_2(x) = \ln|x| J_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n x^{n-|\nu|}$$

$$y_2'(x) = \ln|x| J'_{|\nu|}(x) + x^{-1} J_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n (n-|\nu|) x^{n-|\nu|-1}$$

$$y_2''(x) = \ln|x| J''_{|\nu|}(x) + 2x^{-1} J'_{|\nu|}(x) - x^{-2} J_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n (n-|\nu|)(n-|\nu|-1) x^{n-|\nu|-2}$$

$$x^2 \ln|x| J''_{|\nu|}(x) + 2x J'_{|\nu|}(x) - J_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n (n-|\nu|)(n-|\nu|-1) x^{n-|\nu|} + x \ln|x| J'_{|\nu|}(x) + J_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n (n-|\nu|) x^{n-|\nu|} + x^2 \ln$$

$$2x J'_{|\nu|}(x) + B_1(1 - 2|\nu|) x^{1-|\nu|} + \sum_{n=2}^{\infty} (B_n n(n - 2|\nu|) + B_{n-2}) x^{n-|\nu|} = 0$$

$$J'_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (2m + |\nu|)}{2m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|-1}$$

$$2x J'_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m 2(2m + |\nu|)}{m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|}$$

$$\sum_{m=0}^{\infty} \frac{(-1)^m 2(2m + |\nu|)}{m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|} + B_1(1 - 2|\nu|) x^{1-|\nu|} + \sum_{n=2}^{\infty} (B_n n(n - 2|\nu|) + B_{n-2}) x^{n-|\nu|} = 0$$

TODO How this = Y □

$$J_{\frac{1}{2}}(x) = \frac{2}{\pi x} \sin(x)$$

$$J_{-\frac{1}{2}}(x) = \frac{2}{\pi x} \cos(x)$$

$$Y_{\frac{1}{2}}(x) = -\frac{2}{\pi x} \cos(x)$$

$$Y_{-\frac{1}{2}}(x) = \frac{2}{\pi x} \sin(x)$$

Proof.

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\Gamma(m+\frac{1}{2}+1) = \prod_{i=0}^{m} \frac{2i+1}{2}\sqrt{\pi} = \frac{(2m+1)!}{2^{2m+1}m!}\sqrt{\pi}$$

$$\begin{aligned} J_{\frac{1}{2}}(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+\frac{1}{2}+1)}(\frac{x}{2})^{2m+\frac{1}{2}} \\ &= \sum_{m=0}^{\infty} \frac{(-1)^m 2^{2m+1}\sqrt{\pi}}{(2m+1)!}(\frac{x}{2})^{2m+\frac{1}{2}} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1} \\ &= \sqrt{\frac{2}{\pi x}} \sin(x) \end{aligned}$$

$$\Gamma(m-\frac{1}{2}+1) = \prod_{i=1}^{m} \frac{2i-1}{2}\sqrt{\pi} = \frac{(2m)!}{2^{2m}m!}\sqrt{\pi}$$

$$\begin{aligned} J_{-\frac{1}{2}}(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m-\frac{1}{2}+1)}(\frac{x}{2})^{2m-\frac{1}{2}} \\ &= \sum_{m=0}^{\infty} \frac{(-1)^m 2^{2m}\sqrt{\pi}}{(2m)!}(\frac{x}{2})^{2m-\frac{1}{2}} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \cos(x) \end{aligned}$$

$$Y_{\frac{1}{2}}(x) = \frac{J_{\frac{1}{2}}(x)\cos\left(\frac{1}{2}\pi\right) - J_{-\frac{1}{2}}(x)}{\sin\left(\frac{1}{2}\pi\right)} = -J_{-\frac{1}{2}}(x) = -\frac{2}{\pi x}\cos(x)$$

$$Y_{-\frac{1}{2}}(x) = \frac{J_{-\frac{1}{2}}(x)\cos\left(-\frac{1}{2}\pi\right) - J_{\frac{1}{2}}(x)}{\sin\left(-\frac{1}{2}\pi\right)} = J_{\frac{1}{2}}(x) = \frac{2}{\pi x}\sin(x)$$

□

ii Modified Bessel functions

Modified Bessel functions are a family of special functions that are solutions to the modified Bessel's equation:

$$x^2 y'' + xy' - (x^2+\nu^2)y = 0,$$

where $x \in \mathbb{R}$ is the independent variable and $\nu \in \mathbb{R}$ is called order.

- Modified Bessel function of the first kind:

$$I_\nu(x) = \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu}.$$

- For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

$$I_\nu(x) = \sum_{m=0}^{\infty} \frac{1}{m!\Gamma(m+\nu+1)} (\frac{x}{2})^{2m+\nu}.$$

  $I_\nu(x)$ and $I_{-\nu}(x)$ are two linearly independent solutions of the modified Bessel's equation. $I_\nu(x)$ is defined on $\mathbb{R}_{>=0}$ for $\nu \in \mathbb{R}_{>0} \setminus \mathbb{Z}$, and on $\mathbb{R}_{>0}$ for $\nu \in \mathbb{R}_{<0} \setminus \mathbb{Z}$. $I_\nu(0) = 0$ for $\nu \in \mathbb{R}_{>0} \setminus \mathbb{Z}$.

- For $\nu \in \mathbb{Z}$:

$$I_\nu(x) = I_{-\nu}(x) = I_\nu(-x) = \sum_{m=0}^{\infty} \lim_{\mu \to \nu} \frac{1}{m!\Gamma(m+\mu+1)} (\frac{x}{2})^{2m+\nu} = \sum_{m=0}^{\infty} \frac{1}{m!(m+|\nu|)!} (\frac{x}{2})^{2m+|\nu|}$$

  $I_\nu(x)$ is defined on $\mathbb{R}$. $I_0(0) = 1$. $I_\nu(0) = 0$ for $\nu \in \mathbb{Z}_{\neq 0}$.

• Modified Bessel function of the second kind:

$$K_\nu(x) = \lim_{\mu \to \nu} \frac{\pi}{2} \frac{I_{-\mu}(x) - I_\mu(x)}{\sin(\mu\pi)}.$$

- For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

$$K_\nu(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{\sin(\nu\pi)},$$

  which is a linear combination of $I_\nu(x)$ and $I_{-\nu}(x)$.

- For $\nu \in \mathbb{Z}$:

$$K_\nu(x) = \lim_{\mu \to \nu} K_\mu(x),$$

  which is a solution that is linearly independent to $I_\nu(x)$. $K_\nu(x)$ is defined on $\mathbb{R}_{>0}$.

$$\lim_{x \to 0^+} K_\nu(x) = \infty.$$

The general solution is:

• For $\nu \in \mathbb{R}$:

$$y(x) = CI_\nu(x) + DK_\nu(x).$$

• For $\nu \in \mathbb{R} \setminus \mathbb{Z}$:

$$y(x) = CI_\nu(x) + DI_{-\nu}(x).$$

Proof.

$$x^2y'' + xy' - (x^2 + \nu^2)y = 0$$

$$y'' + x^{-1}y' - (1 + \nu^2x^{-2})y = 0$$

0 is a regular singular point.

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1)x^{n+r-2}$$

$$\sum_{n=0}^{\infty} a_n (n+r)(n+r-1)x^{n+r} + \sum_{n=0}^{\infty} a_n (n+r)x^{n+r} - \sum_{n=0}^{\infty} a_n x^{n+r+2} - \nu^2 \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$L = a_0 r(r-1)x^r + a_0 r x^r - a_0 x^{r+2} - \nu^2 a_0 x^r$$

$$I(r) = r^2 - \nu^2 = 0$$

$$r = \pm\nu$$

$$y_1(x) = \sum_{n=0}^{\infty} A_n x^{n+|\nu|}$$

$$y_1'(x) = \sum_{n=0}^{\infty} A_n (n+|\nu|)x^{n+|\nu|-1}$$

$$y_1''(x) = \sum_{n=0}^{\infty} A_n (n+|\nu|)(n+|\nu|-1)x^{n+|\nu|-2}$$

$$y_2(x) = \sum_{n=0}^{\infty} B_n x^{n-|\nu|}$$

$$y_2'(x) = \sum_{n=0}^{\infty} B_n (n-|\nu|)x^{n-|\nu|-1}$$

$$y_2''(x) = \sum_{n=0}^{\infty} B_n (n-|\nu|)(n-|\nu|-1)x^{n-|\nu|-2}$$

$$\sum_{n=0}^{\infty} A_n (n+|\nu|)(n+|\nu|-1)x^{n+|\nu|} + \sum_{n=0}^{\infty} A_n (n+|\nu|)x^{n+|\nu|} - \sum_{n=0}^{\infty} A_n x^{n+|\nu|+2} - |\nu|^2 \sum_{n=0}^{\infty} A_n x^{n+|\nu|} = 0$$

$$A_1(1+2|\nu|)x^{1+|\nu|} + \sum_{n=2}^{\infty} (A_n(n^2+2n|\nu|) - A_{n-2})x^{n+|\nu|} = 0$$

$$A_n = 0, \quad \frac{n+1}{2} \in \mathbb{N}$$

$$A_n(n^2+2n|\nu|) - A_{n-2} = 0, \quad \frac{n}{2} \in \mathbb{N}$$

$$A_{2m} = \frac{A_0}{\prod_{i=1}^{m}(2i)(2i+2|\nu|)} = \lim_{\mu\to|\nu|} \frac{A_0\Gamma(\mu+1)}{2^{2m}m!\Gamma(m+\mu+1)}, \quad m \in \mathbb{N}$$

Let

$$A_0 = \lim_{\mu\to|\nu|} \frac{1}{2^{\mu}\Gamma(\mu+1)}$$

$$A_{2m} = \lim_{\mu\to|\nu|} \frac{1}{2^{2m+\mu}m!\Gamma(m+\mu+1)}, \quad m \in \mathbb{N}$$

$$y_1(x) = I_{|\nu|}(x) = \sum_{m=0}^{\infty} \lim_{\mu\to|\nu|} \frac{1}{2^{2m+\mu}m!\Gamma(m+\mu+1)} x^{2m+|\nu|} = \sum_{m=0}^{\infty} \lim_{\mu\to|\nu|} \frac{1}{m!\Gamma(m+\mu+1)} (\frac{x}{2})^{2m+|\nu|}$$

$$y_2(x) = I_{-|\nu|}(x) = \sum_{m=0}^{\infty} \lim_{\mu\to|\nu|} \frac{1}{m!\Gamma(m-\mu+1)} (\frac{x}{2})^{2m-|\nu|}$$

For $\nu \in \mathbb{R} \setminus \mathbb{Z}$, that is,

$$I_\nu(x) = \sum_{m=0}^{\infty} \frac{1}{m!\Gamma(m+\nu+1)}(\frac{x}{2})^{2m+\nu}$$

$$I_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{1}{m!\Gamma(m-\nu+1)}(\frac{x}{2})^{2m-\nu}$$

For $2\nu \in \mathbb{Z}$, test whether they are are linearly dependent.

$$I'_\nu(x) = \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1}\frac{2m+\nu}{2}$$

$$I'_{-\nu}(x) = \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m-\mu+1)}(\frac{x}{2})^{2m-\nu-1}\frac{2m-\nu}{2}$$

$$W(I_\nu, I_{-\nu})(x) = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \lim_{\mu\to\nu} \frac{(m-n-\nu)}{m!n!\Gamma(m-\mu+1)\Gamma(n+\mu+1)}(\frac{x}{2})^{2m+2n-1}$$

The coefficient of the x^{-1} term is

$$\lim_{\mu\to\nu} \frac{-2\nu}{\Gamma(1-\mu)\Gamma(1+\mu)} = \lim_{\mu\to\nu} \frac{-2}{\Gamma(\mu)\Gamma(1-\mu)} = \frac{-2\sin(\pi\nu)}{\pi}$$

Thus for $\nu \in \mathbb{Z}$, $W(I_\nu, I_{-\nu}) = 0$,

$$I_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{1}{m!(m+|\nu|)!}(\frac{x}{2})^{2m+|\nu|}$$

and observe that for $k \in \mathbb{N}_0$,

$$\lim_{\mu\to\nu} \frac{1}{\Gamma(-k)} = 0,$$

thus

$$I_{-|\nu|}(x) = \sum_{m=0}^{\infty} \frac{1}{m!(m+|\nu|)!}(\frac{x}{2})^{2m+|\nu|},$$

and

$$I_{-|\nu|}(x) = I_{|\nu|}(x),$$

that is,

$$I_\nu(x) = I_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{1}{m!(m+|\nu|)!}(\frac{x}{2})^{2m+|\nu|},$$

and a new $y_2(x)$ is

$$y_2(x) = \ln|x| I_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n x^{n-|\nu|}$$

$$y_2'(x) = \ln|x| I'_{|\nu|}(x) + x^{-1}I_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n(n-|\nu|)x^{n-|\nu|-1}$$

$$y_2''(x) = \ln|x| I''_{|\nu|}(x) + 2x^{-1}I'_{|\nu|}(x) - x^{-2}I_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n(n-|\nu|)(n-|\nu|-1)x^{n-|\nu|-2}$$

$$x^2 \ln|x| I''_{|\nu|}(x) + 2xI'_{|\nu|}(x) - I_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n(n-|\nu|)(n-|\nu|-1)x^{n-|\nu|} + x\ln|x| I'_{|\nu|}(x) + I_{|\nu|}(x) + \sum_{n=0}^{\infty} B_n(n-|\nu|)x^{n-|\nu|} - x^2 \ln$$

$$2xI'_{|\nu|}(x) + B_1 x^{1-|\nu|} + \sum_{n=2}^{\infty}(B_n(n^2 - 2n\,|\nu|) - B_{n-2})x^{n-|\nu|} = 0$$

$$I'_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{(2m + |\nu|)}{2m!(m + |\nu|)!}(\frac{x}{2})^{2m+|\nu|-1}$$

$$2xI'_{|\nu|}(x) = \sum_{m=0}^{\infty} \frac{2(2m + |\nu|)}{m!(m + |\nu|)!}(\frac{x}{2})^{2m+|\nu|}$$

$$\sum_{m=0}^{\infty} \frac{2(2m + |\nu|)}{m!(m + |\nu|)!}(\frac{x}{2})^{2m+|\nu|} + B_1 x^{1-|\nu|} + \sum_{n=2}^{\infty}(B_n(n^2 - 2n\,|\nu|) - B_{n-2})x^{n-|\nu|} = 0$$

TODO How this = K □

iii Spherical Bessel functions

Spherical Bessel functions are a family of special functions that are solutions to the equation:

$$x^2 y'' + 2xy' + (x^2 - n(n-1))y = 0,$$

where $x \in \mathbb{R}$ is the independent variable and $n \in \mathbb{Z}$.

- Spherical Bessel function of the first kind:

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x).$$

- Spherical Bessel function of the second kind:

$$y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+\frac{1}{2}}(x) = (-1)^{n+1}\sqrt{\frac{\pi}{2x}} J_{-n-\frac{1}{2}}(x),$$

  which is linearly independent from $j_n(x)$.

$$j_n(x) = (-1)^n y_{-n-1}(x)$$

$$y_n(x) = (-1)^{n+1} j_{-n-1}(x)$$

$$j_0(x) = \frac{\sin(x)}{x}$$

$$y_0(x) = -\frac{\cos(x)}{x}$$

The general solution is:

$$y = Cj_n(x) + Dy_n(x).$$

Proof.

$$u = x^{0.5} y$$

$$u' = x^{0.5} y' + 0.5x^{-0.5} y$$

$$u'' = x^{0.5} y'' + x^{-0.5} y' - 0.25x^{-1.5} y$$

$$x^{1.5} u'' = x^2 y'' + xy' - 0.25y$$

$$x^{0.5} u' = xy' + 0.5y$$

$$x^{1.5}u'' + x^{0.5}u' + (x^2 - n(n-1) - 0.25)x^{-0.5}u = 0,$$

$$x^2u'' + xu' + (x^2 - (n-0.5)^2)u = 0$$

$$u = CJ_{n+0.5}(x) + DY_{n+0.5}(x)$$

□

iv Parametric Bessel equation

ODE:

$$x^2y'' + xy' + (\alpha^2x^2 - \nu^2)y = 0, \quad \alpha \neq 0$$

Solution:

$$y(x) = CJ_\nu(\alpha x) + DY_\nu(\alpha x).$$

Proof. Let

$$z = \alpha x$$

$$y(x) = u(z)$$

$$y'(x) = \alpha u'(z)$$

$$y''(x) = \alpha^2 u''(z)$$

$$z^2u'' + zu' + (z^2 - \nu^2)u = 0$$

$$u(z) = CJ_\nu(z) + DY_\nu(z).$$

□

v Parametric Modified Bessel equation

ODE:

$$x^2y'' + xy' - (\alpha^2x^2 + \nu^2)y = 0, \quad \alpha \neq 0$$

Solution:

$$y(x) = CI_\nu(\alpha x) + DK_\nu(\alpha x).$$

Proof. Let

$$z = \alpha x$$

$$y(x) = u(z)$$

$$y'(x) = \alpha u'(z)$$

$$y''(x) = \alpha^2 u''(z)$$

$$z^2u'' + zu' - (z^2 + \nu^2)u = 0$$

$$u(z) = CI_\nu(z) + DK_\nu(z).$$

□

vi Derivative of Bessel function of first kind

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^{-\nu}J_\nu(x)) = -x^{-\nu}J_{\nu+1}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^{-\nu}J_\nu(x)) &= -\nu x^{-\nu-1}J_\nu(x) + x^{-\nu}J_\nu'(x) \\
&= -\nu x^{-\nu-1}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + x^{-\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= -x^{-\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m\nu}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + x^{-\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{-\nu}\sum_{m=1}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{-\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^{m+1}}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1} \\
&= -x^{-\nu}J_{\nu+1}(x)
\end{aligned}$$

□

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^{\nu}J_\nu(x)) = x^{\nu}J_{\nu-1}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^{\nu}J_\nu(x)) &= \nu x^{\nu-1}J_\nu(x) + x^{\nu}J_\nu'(x) \\
&= \nu x^{\nu-1}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + x^{\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m\nu}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + x^{\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu}\sum_{m=0}^{\infty}\lim_{\mu\to\nu}\frac{(-1)^m}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu}J_{\nu-1}(x)
\end{aligned}$$

□

$$xJ_\nu'(x) = \nu J_\nu(x) - xJ_{\nu+1}(x).$$

Proof.

$$
\begin{aligned}
xJ_\nu'(x) &= \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= \nu \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + 2\sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= \nu J_\nu(x) + x\sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= \nu J_\nu(x) + x\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^{m+1}}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1} \\
&= \nu J_\nu(x) - xJ_{\nu+1}(x)
\end{aligned}
$$

□

$$xJ_\nu'(x) = xJ_{\nu-1}(x) - \nu J_\nu(x).$$

Proof.

$$
\begin{aligned}
xJ_\nu'(x) &= \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= 2\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m(m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} - \nu\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= x\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} - \nu\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= xJ_{\nu-1}(x) - \nu J_\nu(x)
\end{aligned}
$$

□

$$J_\nu'(x) = \frac{1}{2}(J_{\nu-1}(x) - J_{\nu+1}(x)).$$

Proof.

$$
\begin{aligned}
J_\nu'(x) &= \frac{1}{2}\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= \frac{1}{2}(\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m(m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1}) \\
&= \frac{1}{2}(\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^m}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} + \sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{(-1)^{m+1}}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1}) \\
&= \frac{1}{2}(J_{\nu-1}(x) - J_{\nu+1}(x))
\end{aligned}
$$

□

vii Derivative of modified Bessel function of first kind

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^{-\nu} I_\nu(x)) = x^{-\nu} I_{\nu+1}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^{-\nu} I_\nu(x)) &= -\nu x^{-\nu-1} I_\nu(x) + x^{-\nu} I_\nu'(x) \\
&= -\nu x^{-\nu-1} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + x^{-\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= -x^{-\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{\nu}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + x^{-\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{-\nu} \sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{1}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{-\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1} \\
&= x^{-\nu} I_{\nu+1}(x)
\end{aligned}$$

□

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^{\nu} I_\nu(x)) = x^{\nu} I_{\nu-1}(x).$$

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^{\nu} I_\nu(x)) &= \nu x^{\nu-1} I_\nu(x) + x^{\nu} I_\nu'(x) \\
&= \nu x^{\nu-1} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + x^{\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{\nu}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + x^{\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{2m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu} \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} \\
&= x^{\nu} I_{\nu-1}(x)
\end{aligned}$$

□

$$x I_\nu'(x) = \nu I_\nu(x) + x I_{\nu+1}(x).$$

Proof.

$$
\begin{aligned}
xI_\nu'(x) &= \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= \nu \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} + 2\sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{1}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= \nu I_\nu(x) + x\sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{1}{(m-1)!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= \nu I_\nu(x) + x\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1} \\
&= \nu I_\nu(x) + xI_{\nu+1}(x)
\end{aligned}
$$

□

$$xI_\nu'(x) = xI_{\nu-1}(x) - \nu I_\nu(x).$$

Proof.

$$
\begin{aligned}
xI_\nu'(x) &= \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= 2\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} - \nu\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= x\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} - \nu\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu} \\
&= xI_{\nu-1}(x) - \nu I_\nu(x)
\end{aligned}
$$

□

$$I_\nu'(x) = \frac{1}{2}(I_{\nu-1}(x) + I_{\nu+1}(x)).$$

Proof.

$$
\begin{aligned}
I_\nu'(x) &= \frac{1}{2}\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(2m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} \\
&= \frac{1}{2}(\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{(m+\nu)}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1} + \sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{m}{m!\Gamma(m+\mu+1)}(\frac{x}{2})^{2m+\nu-1}) \\
&= \frac{1}{2}(\sum_{m=0}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu)}(\frac{x}{2})^{2m+\nu-1} + \sum_{m=1}^{\infty} \lim_{\mu\to\nu} \frac{1}{m!\Gamma(m+\mu+2)}(\frac{x}{2})^{2m+\nu+1}) \\
&= \frac{1}{2}(I_{\nu-1}(x) + I_{\nu+1}(x))
\end{aligned}
$$

□

viii Legendre polynomial

Legendre polynomials are the polynomial solutions of Legendre's differential equations for $|x| < 1$:

$$(1-x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = 0,$$

where $n \in \mathbb{N}_0$.

$$P_n(x) = \frac{1}{2^n}\sum_{k=0}^{\left\lfloor \frac{n}{2} \right\rfloor}(-1)^k\binom{2n-2k}{n}\binom{n}{k}x^{n-2k}$$

Proof.

$$P_n''(x) - \frac{2x}{1-x^2}P_n'(x) + \frac{n(n+1)}{1-x^2}P_n(x) = 0.$$

± 1 are two regular singular points.

Use power series method at 0.

$$y(x) = \sum_{m=0}^{\infty} a_m x^m$$

$$y'(x) = \sum_{m=1}^{\infty} a_m m x^{m-1}$$

$$y''(x) = \sum_{m=2}^{\infty} a_m m(m-1)x^{m-2}$$

$$\sum_{m=2}^{\infty} a_m m(m-1)x^{m-2} - \sum_{m=2}^{\infty} a_m m(m-1)x^m - 2\sum_{m=1}^{\infty} a_m m x^m + n(n+1)\sum_{m=0}^{\infty} a_m x^m = 0$$

$$2a_2 + n(n+1)a_0 + (6a_3 - 2a_1 + n(n+1)a_1)x + \sum_{m=2}^{\infty}(a_{m+2}(m+2)(m+1) - a_m(m+1) + n(n+1)a_m)x^m = 0$$

$$a_2 = -\frac{n(n+1)}{2}a_0$$

$$a_3 = -\frac{(n-1)(n+2)}{6}a_1$$

$$a_{m+2} = -\frac{(n-m)(n+m+1)}{(m+1)(m+2)}a_m, \quad m \in \mathbb{N}_0$$

$$a_m = -\frac{(m+1)(m+2)}{(n-m)(n+m+1)}a_{m+2}, \quad m \in \mathbb{N}_0$$

a_m terminates at $m = n$, which yields the polynomial solution.

For even n, let $c = \frac{n}{2}$.

$$a_n = (-1)^c\frac{(2n)!(c!)^2}{(n!)^3}a_0$$

Take

$$a_0 = (-1)^c\frac{n!}{2^n(c!)^2}$$

$$a_n = \frac{(2n)!}{2^n(n!)^2}$$

$$a_{n-2k} = (-1)^k \frac{n!}{(n-2k)!} \frac{1}{2^k k!} \frac{2^n n!(2n-2k)!}{2^{n-k}(n-k)!(2n)!} a_n$$

$$a_{n-2k} = (-1)^k \frac{(2n-2k)!}{(n-2k)!} \frac{1}{k!(n-k)!} \frac{1}{2^n} = \frac{(-1)^k}{2^n} \binom{2n-2k}{n} \binom{n}{k}$$

For odd n, let $c = \dfrac{n-1}{2}$.

$$a_n = (-1)^c \frac{(2n)!(c!)^2}{2(n!)^3} a_1$$

Take

$$a_1 = (-1)^c \frac{n!}{2^{n-1}(c!)^2}$$

$$a_n = \frac{(2n)!}{2^n (n!)^2}$$

$$a_{n-2k} = (-1)^k \frac{n!}{(n-2k)!} \frac{1}{2^k k!} \frac{2^n n!(2n-2k)!}{2^{n-k}(n-k)!(2n)!} a_n$$

$$a_{n-2k} = (-1)^k \frac{(2n-2k)!}{(n-2k)!} \frac{1}{k!(n-k)!} \frac{1}{2^n} = \frac{(-1)^k}{2^n} \binom{2n-2k}{n} \binom{n}{k}$$

Thus, for all $n \in \mathbb{N}_0$,

$$P_n(x) = \frac{1}{2^n} \sum_{k=0}^{\left\lfloor \frac{n}{2} \right\rfloor} (-1)^k \binom{2n-2k}{n} \binom{n}{k} x^{n-2k}$$

□

ix Rodrigues' formula

The Legendre polynomials are given by

$$P_n(x) = \frac{1}{2^n n!} \frac{\mathrm{d}^n}{\mathrm{d}x^n} (x^2 - 1)^n$$

Proof. TODO □

4 Linear First-order System

I Basics

i System of linear first-order ODEs or linear first-order system or linear system

A linear first-order system can be written in matrix form

$$\mathbf{y}'(x) = \mathbf{A}(x)\mathbf{y}(x) + \mathbf{f}(x),$$

where

$$\mathbf{y}(x) = \begin{bmatrix} y_1(x) \\ \vdots \\ y_n(x) \end{bmatrix}$$

is called solution vector,

$$\mathbf{A}(x) = \begin{bmatrix} a_{1,1}(x) & \cdots & a_{1,n}(x) \\ \vdots & \ddots & \vdots \\ a_{n,1}(x) & \cdots & a_{n,n}(x) \end{bmatrix}$$

is called coefficient matrix, and

$$\mathbf{f}(x) = \begin{bmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{bmatrix}$$

is called forcing term.

And an IC of an IVP of it can be written in the form

$$\mathbf{y}(x_0) = \mathbf{y}_0 = \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_n \end{bmatrix}$$

where x_0 is a constant in the domain of **A** and **f**, and $\mathbf{y}_0$ is a constant vector.

A homogeneous one is called homogeneous system.

ii Solving by Elimination

1. Write the system in terms of differential operator D.
2. Manipulate the polynomials of D and unknowns such that each linear equation contains only each unknown.
3. Solve all of them.

iii Example 1

ODEs:

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\mathrm{d}y}{\mathrm{d}t} + 2y = 0$$
$$\frac{\mathrm{d}x}{\mathrm{d}t} - 3x - 2y = 0$$

Solutions:

$$x(t) = Ce^{2t} + De^{-3t}$$
$$y(t) = Ae^{2t} + Be^{-3t}$$

Proof.

$$Dx + (D+2)y = 0$$
$$(D-3)x - 2y = 0$$
$$(D-3)Dx + (D-3)(D+2)y = 0$$
$$D(D-3)x - D2y = 0$$
$$(D^2 + D - 6)y = 0$$
$$r = 2 \vee -3$$

$$y(t) = Ae^{2t} + Be^{-3t}$$

$$2Dx + 2(D+2)y = 0$$

$$(D+2)(D-3)x - (D+2)2y = 0$$

$$(D^2 + D - 6)x = 0$$

$$x(t) = Ce^{2t} + De^{-3t}$$

□

iv Wrońskian

Suppose n functions $\mathbf{y}_i : \Omega_i \subseteq \mathbb{R} \to \mathbb{R}^n$ for all $i \in \mathbb{N}_{\leq n}$ are solution vectors to a homogeneous system on an interval and

$$\mathbf{y}_i = \begin{bmatrix} y_{1i} \\ \vdots \\ y_{ni} \end{bmatrix}.$$

Then the Wrońskian of them is defined as

$$W(\mathbf{y}_1, \dots, \mathbf{y}_n) = \begin{vmatrix} \mathbf{y}_1 & \dots & \mathbf{y}_n \end{vmatrix} = \begin{vmatrix} y_{11} & \cdots & y_{1n} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{nn} \end{vmatrix}.$$

v Non-zero Wrońskian implies linear independence

For n functions $\mathbf{y}_i : \Omega_i \subseteq \mathbb{R} \to \mathbb{R}^n$ for all $i \in \mathbb{N}_{\leq n}$, if $W(\mathbf{y}_1, \dots, \mathbf{y}_n) \neq 0$, then they are linearly independent.

Proof. If they are not linearly independent, there exists not all zero $c_1, \dots, c_n$ such that

$$\sum_{i=1}^{n} c_i \mathbf{y}_i = 0,$$

tha is,

$$\begin{bmatrix} y_{11} & \cdots & y_{1n} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{nn} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

All $c_i = 0$ if $W(\mathbf{y}_1, \dots, \mathbf{y}_n) \neq 0$ by Cramer rule, thus $W(\mathbf{y}_1, \dots, \mathbf{y}_n) = 0$. □

vi Homogeneity

A linear first-order system is called a homogeneous linear first-order system iff the forcing term is 0 for all x; otherwise, it is called a non-homogeneous linear first-order system.

vii Complementary or associated (homogeneous) function, natural, complementary, or homogeneous solution or function, and particularly solution

Given an nth-order linear first-order system

$$\mathbf{y}'(x) = \mathbf{A}(x)\mathbf{y}(x) + \mathbf{f}(x),$$

$$\mathbf{y}'(x) = \mathbf{A}(x)\mathbf{y}(x),$$

is called its complementary or associated (homogeneous) function, and the general solutions of the original equation is the sum of the general solutions $\mathbf{y}_c$ of the complementary function, called natural, complementary, or homogeneous solution or function, and a particular solution $\mathbf{y}_p$.

viii Superposition principle

If $\mathbf{y}_1(x)$ and $\mathbf{y}_2(x)$ are both solutions to a linear first-order system on an interval, then $c_1\mathbf{y}_1(x) + c_2\mathbf{y}_2(x)$ is also a solution to it on the interval for any $c_1, c_2 \in \mathbb{R}$.

II Homogeneous linear first-order system

TODO

5 Classical theory

TODO : below waiting for change to "Ordinary Differential Equations and Dynamical Systems" version. : above things e.g. solutions as vector space etc. mb need conditions or assumptions of existence and/or uniqueness.

I Existence and Uniqueness of Solution

i Peano existence theorem (皮亞諾存在性定理), Peano theorem (皮亞諾定理), or Cauchy–Peano theorem (柯西-皮亞諾定理)

Let D be an open subset of $\mathbb{R} \times \mathbb{R}^n$ and $f : D \to \mathbb{R}$ be a continuous function, then every initial value problem given by explicit first-order ODE $y'(t) = f(t, y(t))$ defined on D and initial condition $y\left(t_0\right) = y_0$ with $(t_0, y_0) \in D$ has a local solution $z : I \to \mathbb{R}^n$ where I is a neighbourhood of t_0 in $\mathbb{R}$.

ii Carathéodory's existence theorem

Let

$$D = \{(t, y) \mid |t - T| \leq a \wedge \forall \text{ integer } 1 \leq i \leq n : |y_i - Y_i| \leq b_i\}$$

be a rectangle in $\mathbb{R} \times \mathbb{R}^n$ where $t, T \in \mathbb{R}$, $y = (y_1, y_2, \dots y_n) \in \mathbb{R}^n$, $Y = (Y_1, Y_2, \dots Y_n) \in \mathbb{R}^n$, and constant $a, b_i \in \mathbb{R}_{>0}$, and $f(t, y) : D \to \mathbb{R}$ be a function that is:

- continuous in y for each fixed t,
- Lebesgue-measurable in t for each fixed y, and

- such that there is a Lebesgue-integrable function $m\colon [T-a, T+a] \to [0,\infty)$ such that $|f(t,y)| \le m(t)$ for all $(t,y) \in D$,

then every initial value problem given by explicit first-order ODE $y'(t) = f(t, y(t))$ defined on D and initial condition $y(t_0) = y_0$ with $(t_0, y_0) \in D$ has a local solution $z\colon I \to \mathbb{R}^n$ where I is a neighbourhood of t_0 in $\mathbb{R}$.

Let I be an open interval of $\mathbb{R}$ and $f(t,y)\colon I \times \mathbb{R}^n \to \mathbb{R}$ where $t \in \mathbb{R}$, $y = (y_1, y_2, \dots y_n) \in \mathbb{R}^n$ be a function that is:

- continuous in y for each fixed t,
- Lebesgue-measurable in t for each fixed y, and
- such that there is a Lebesgue-integrable function $m\colon I \to [0,\infty)$ such that $|f(t,y)| \le m(t)$ for all $(t,y) \in I \times \mathbb{R}^n$,

then every initial value problem given by explicit first-order ODE $y'(t) = f(t, y(t))$ defined on $I \times \mathbb{R}^n$ and initial condition $y(t_0) = y_0$ with $(t_0, y_0) \in I \times \mathbb{R}^n$ has a local solution $z\colon I \to \mathbb{R}^n$ where I is a neighbourhood of t_0 in $\mathbb{R}$.

iii Picard–Lindelöf theorem (皮卡-林德勒夫定理) or Cauchy–Lipschitz theorem (柯西-利普希茨定理) local version

Let

$$D = \{(t,y) \mid |t-T| \le a \land \forall \text{ integer } 1 \le i \le n\colon |y_i - Y_i| \le b_i\}$$

be a rectangle in $\mathbb{R} \times \mathbb{R}^n$ where $t, T \in \mathbb{R}$, $y = (y_1, y_2, \dots y_n) \in \mathbb{R}^n$, $Y = (Y_1, Y_2, \dots Y_n) \in \mathbb{R}^n$, and constant $a, b_i \in \mathbb{R}_{>0}$, and $f(t,y)\colon D \to \mathbb{R}$ be a function that is continuous in t and locally Lipschitz on D in all y_i, then every initial value problem given by explicit first-order ODE $y'(t) = f(t, y(t))$ defined on D and initial condition $y(t_0) = y_0$ with $(t_0, y_0) \in D$ has a unique local solution $z\colon I \to \mathbb{R}^n$ where I is a neighbourhood of t_0 in $\mathbb{R}$, that is, let I_1, I_2 be two neighbourhoods of t_0 in $\mathbb{R}$ and $z_i\colon I_i \to \mathbb{R}^n$ for $i \in \{1,2\}$ be two differentiable functions such that $z_i'(t) = f\left(t, z_i(t)\right)$ for all $t \in I_i$, for $i \in \{1,2\}$, then $\exists t_0 \in I_1 \cap I_2$ s.t. $z_1(t_0) = z_2(t_0) \implies \forall t \in I_1 \cap I_2\colon z_1(t) = z_2(t)$.

Let I be an open interval of $\mathbb{R}$ and $f(t,y)\colon I \times \mathbb{R}^n \to \mathbb{R}$ where $t \in \mathbb{R}$, $y = (y_1, y_2, \dots y_n) \in \mathbb{R}^n$ be a function that is continuous in t and locally Lipschitz in all y_i, then every initial value problem given by explicit first-order ODE $y'(t) = f(t, y(t))$ defined on $I \times \mathbb{R}^n$ and initial condition $y(t_0) = y_0$ with $(t_0, y_0) \in I \times \mathbb{R}^n$ has a unique solution $z\colon I \to \mathbb{R}^n$, that is, let $I_1, I_2 \subseteq \mathbb{R}$ and $z_i\colon I_i \to \mathbb{R}^n$ for $i \in \{1,2\}$ be two differentiable functions such that $z_i'(t) = f\left(t, z_i(t)\right)$ for all $t \in I_i$, for $i \in \{1,2\}$, then $\exists t_0 \in I_1 \cap I_2$ s.t. $z_1(t_0) = z_2(t_0) \implies \forall t \in I_1 \cap I_2\colon z_1(t) = z_2(t)$.

iv Existence of a unique solution of linear ODE

Let $n \in \mathbb{Z}$, $a_i(x)$ for all $i \in [0,n] \cap \mathbb{Z}$ and $g(x)$ be continuous on an interval I, $a_n(x) \neq 0$ on I, and $x_0 \in I$. Then the IVP of

$$\sum_{i=0}^{n} a_i(x) \frac{\mathrm{d}^i}{\mathrm{d}(^i} y)(x) = g(x)$$

subjected to $y^{(i)}(x_0) = y_i$ for all $i \in [0, n-1] \cap \mathbb{Z}$ has a solution $y(x)$ on I and the solution is unique. TODO: BVP can have many, one, or no solution TODO: end

6 Dynamical system (動態系統 or 動力系統)

I Dynamical system (DS)

i Dynamical system

A dynamical system, or simply system, is a tuple (T, X, Φ) where T is a monoid, written additively and with the independent variable t in it called evolution parameter (演化參數) or time, X is a non-empty set, called the phase space (相空間) or state space (狀態空間) with the independent variable x in it called (system) phase (相) or state (狀態), and Φ is a function, called the evolution function (演化函數) or flow (map),

$$\Phi(t, x)\colon\ U \subseteq (T \times X) \to X$$

where U is such that the second projection map π_2 for $T \times X$ satisfies

$$\pi_2(U) = X,$$

and for any $x \in X$, $\Phi(0, x) = x$.

ii Time-invariant or autonomous dynamical system

A dynamic system (T, X, Φ) is called time-invariant or autonomous iff it satisfies the semigroup property, translation property, shift property, time-invariance, or autonomy that

$$\Phi(t_2, \Phi(t_1, x)) = \Phi(t_2 + t_1, x)$$

for any $t_1, t_2 + t_1 \in I(x)$ and $t_2 \in I(\Phi(t_1, x))$, in which

$$I(x) \coloneqq \{t \in T \mid (t, x) \in U\}$$

for any $x \in X$.

Otherwise, it is called time-variant or non-autonomous.

iii Orbit (軌道)

Given a dynamical system (T, X, Φ) with

$$\Phi(t, x)\colon\ U \subseteq (T \times X) \to X$$

and

$$I(x) \coloneqq \{t \in T \mid (t, x) \in U\}.$$

The set

$$\gamma(x_0) \coloneqq \{\Phi(t, x)\colon\ t \in I(x)\} \subset X$$

is called the orbit through x.

An orbit which consists of a single point is called constant orbit. A non-constant orbit is called closed or periodic if there exists $t \neq 0 \in I(x)$ such that $\Phi(t, x) = x$.

iv (Phase) trajectory (function)

Given a dynamical system (T, X, Φ), for any $x_0 \in X$, the (phase) trajectory (function) starting at x_0 is a function

$$x : t \to \gamma(x_0);\ t \mapsto \Phi(t, x_0).$$

v Phase portrait (相圖)

A phase portrait is a geometric representation of a collection of phase trajectories of an autonomous dynamical system, typically all phase trajectories starting at any $x \in X$.

vi Attractor (吸引子)

For a dynamical system $(\mathbb{R}, X, \Phi)$ with $D \subseteq X$ and $\Phi(t, x) : \mathbb{R}_{\geq 0} \times D \to X$, a non-empty subset A of D is called an attractor if and only if it satisfies the following three conditions:

- Forward invariance:
$$\forall a \in A : \forall t > 0 : \Phi(t, a) \in A.$$
- Attraction: There exists a neighborhood of A, called the basin of attraction for A and denoted $B(A)$, such that for any $b \in B(A)$ and any open neighborhood N of A, there is a positive constant T such that $\forall t > T : \Phi(t, b) \in N$.
- Minimality: There is no non-empty proper subset of A that satisfies the above two conditions.

A point a in D is called an attractor iff $\{a\}$ is an attractor.

vii Repeller

For a dynamical system $(\mathbb{R}, X, \Phi)$ with $D \subseteq X$ and $\Phi(t, x) : \mathbb{R}_{\geq 0} \times D \to X$, a non-empty subset R of X is called a repeller if and only if it satisfies the following three conditions:

- Forward invariance:
$$\forall r \in R : \forall t > 0 : \Phi(t, r) \in R.$$
- Repulsion: There exists a neighborhood of R, called the basin of repulsion for R and denoted $B(R)$, such that for any $b \in B(R) \setminus R$ and any open neighborhood N of R, there is a positive constant T such that $\forall t > T : \Phi(t, b) \notin N$.
- Minimality: There is no non-empty proper subset of R that satisfies the above two conditions.

A point r in D is called a repeller iff $\{r\}$ is a repeller.

viii Continuous-time dynamical system

A continuous-time dynamical system is a dynamical system $(\mathbb{R}, X, \Phi)$ with X being a manifold locally diffeomorphic to a Banach space and $\Phi(t, x) : A \times X \to X$ with $\mathbb{R}_{\geq 0} \subseteq A \subseteq \mathbb{R}$ such that $\frac{\partial \Phi}{\partial t}\Big|_{t=0}$ exists for all $x \in X$.

ix Discrete-time dynamical system

A discrete-time dynamical system is a dynamical system $(\mathbb{Z}, X, \Phi)$ with X being a manifold locally diffeomorphic to a Banach space and $\Phi(t, x) : A \times X \to X$ with $\mathbb{N}_0 \subseteq A \subseteq \mathbb{Z}$.

II Continuous-time dynamical systems

Below, we will discuss continuous-time dynamical systems in Euclidean vector spaces, that are, continuous-time dynamical systems whose phase space is $\mathbb{R}^n$ for some $n \in \mathbb{N}$.

i Systems of first-order ODEs

Consider a system of first-order ODEs:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(t, x), \quad x \in \mathbb{R}^n \wedge f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n.$$

Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be the set such that for any (t_0, x_0) in U, t_0 is in the interval of definition of the solution of it with the initial condition $x(0) = x_0$ and that the solution is unique, and $x(t; x_0)$ be the solution of it with the initial condition $x(0) = x_0$.

Then the dynamical system represented by it has phase space $\mathbb{R}^n$, evolution parameter t, phase x, and evolution function $\Phi : U \to \mathbb{R}^n; \Phi(t, x_0) = x(t; x_0)$. The system of ODEs is called the evolution rule of the dynamical system, and the dynamical system is called to be defined by the system of ODEs. Each solution of the system of ODEs is a phase trajectory of the dynamical system.

III Autonomous continuous-time dynamical systems

Below, we will discuss autonomous continuous-time dynamical systems in Euclidean vector spaces, that are, autonomous continuous-time dynamical systems whose phase space is $\mathbb{R}^n$ for some $n \in \mathbb{N}$.

i Equilibria, fixed points, or fixpoints of autonomous continuous-time dynamical systems

An equilibrium, fixed point, or fixpoint of an autonomous continuous-time dynamical system (T, X, Φ) is a point $x^* \in X$ such that the vector field $f(x) = \frac{\partial \Phi}{\partial t}\Big|_{t=0}$ vanishes there, that is, $f(x^*) = 0$.

ii Autonomous system of ODEs

An autonomous system of ODEs defines an autonomous continuous-time dynamical system, and the equilibria of the dynamical system are the critical points of the system of ODEs.

iii Phase portrait

For an autonomous continuous-time dynamical system, one usually plots X, marks equilibria, and draw arrows at each point x in a grid indicating $f(x)$ at it. When $X = \mathbb{R}$, a phase potrait is also called a phase line.

iv Linear autonomous continuous-time dynamical system

An autonomous continuous-time dynamical system $(\mathbb{R}, \mathbb{R}^n, \Phi(t, x))$ is called linear iff its evolution rule is a system of linear ODEs. Let the forcing term be $q(x) = Cx^\top$ where C is a constant matrix in $\mathbb{R}^{n\times n}$, then C is called the dynamics matrix of the system, and the eigenvalues and eigen vectors of C are called the eigenvalues and eigen vectors of the system.

IV Stability of equilibria

Let x^* be an equilibrium of an autonomous continuous-time dynamical system and A be the domain of the evolution function.

i (Lyapunov) stable (（李亞普諾夫）穩定)

x^* is called to be (Lyapunov) stable iff

$$\forall \varepsilon > 0 :\ \exists \delta > 0 \text{ s.t. } \|x(0) - x^*\| < \delta \implies \forall t \in A :\ \|x(t) - x^*\| < \varepsilon.$$

ii Asymptotically stable (漸近穩定)

x^* is called to be asymptotically stable iff

$$\exists \delta > 0 \text{ s.t. } \|x(0) - x^*\| < \delta \implies \lim_{t\to\infty} x(t) = x^*.$$

An asymptotically stable equilibrium is necessarily (Lyapunov) stable and an attractor.

iii Exponentially stable (指數穩定)

x^* is called to be exponentially stable iff

$$\exists M > 0, \alpha > 0, \delta > 0 \text{ s.t. } \|x(0) - x^*\| < \delta \implies \forall t \in A\|x(t) - x^*\| \le \|x(0) - x^*\|e^{-\alpha t}.$$

An exponentially stable equilibrium is necessarily asymptotically stable.

iv Semi-stable or semistable (半穩定)

x^* is called to be semi-stable or semi-stable iff it is (Lyapunov) stable and

$$(\exists \delta > 0 \text{ s.t. } x(0) - x^* < \delta \implies \lim_{t\to\infty} x(t) = x^*) \vee (\exists \delta > 0 \text{ s.t. } x^* - x(0) < \delta \implies \lim_{t\to\infty} x(t) = x^*).$$

v Marginally stable (臨界穩定)

For linear autonomous continuous-time dynamical system, x^* is called to be marginally stable iff it is (Lyapunov) stable but not asymptotically stable.

vi Unstable (不穩定)

x^* is called to be unstable iff it is not (Lyapunov) stable.

An equilibrium that is in a repeller is necessarily unstable.

7 Numerical Method

I First-order ODE

i (Forward) Euler('s) method

Approximate values for the solution of the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, with step size h, at $x_n = x_{n-1} + h$, are

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1}), \quad n \in \mathbb{N}.$$

ii Backward Euler('s) method

Approximate values for the solution of the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, with step size h, at $x_n = x_{n-1} + h$, are the solutions of the equations

$$y_n = y_{n-1} + hf(x_n, y_n), \quad n \in \mathbb{N}.$$

8 Control Theory (控制理論)

I Continuous-time linear time-invariant (LTI) system

i Causal (因果的) continuous-time linear time-invariant (LTI) system

A causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation, given initial condition $y(0) = 0$. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$ and is such that $h(t) = 0$ for all $t < 0$, that

$$y(t) = (x * h)(t),$$

where

$$(x * h)(t) := \int_0^t x(\tau)h(t - \tau)\,\mathrm{d}\tau\,.$$

Applying unilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is called transfer function (轉移函數), and its absolute value is called gain, or attenuation if < 1.

ii Non-causal continuous-time linear time-invariant (LTI) system

A non-causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$, that

$$y(t) = (x * h)(t) := \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau\,.$$

Applying bilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is called transfer function, and its absolute value is called gain (增益), or attenuation (衰減量) if < 1.

iii Asymptotically stable or bounded-input bounded-output (BIBO) stable

A LTI system is called asymptotically stable or bounded-input bounded-output stable iff bounded input implies bounded response, iff all poles of it are on the left half-plane iff $h(t)$ is absolutely integrable.

iv (Lyapunov) stable

A LTI system is called (Lyapunov) stable iff there is no pole of it on the right half-plane and no multiple pole of it on the imaginary axis iff $h(t)$ is bounded.

v Marginally stable

A LTI system is called marginally stable iff it is stable but not asymptotically stable.

vi Unstable

A LTI system is called unstable iff it is not stable.

vii Transient response

A response component that converges to an equilibrium as t gets large. Such converged state is called steady state or transient state.

viii Time constant

For a response function in the form $y(t) = y_0 e^{-\alpha t}$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\dfrac{1}{\alpha}$, that is, $y(\tau) = y_0 e^{-1}$.

For a response function in the form $y(t) = y_0(1 - e^{-\alpha t})$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\dfrac{1}{\alpha}$, that is, $y(\tau) = y_0(1 - e^{-1})$.

ix Frequency response

When the input is $x(t) = e^{i\omega t}$, $X(s) = \dfrac{1}{s - i\omega}$, the response is

$$Y(s) = \frac{H(s)}{s - i\omega}.$$

We assume the system is stable, that is, all poles $a_1, \dots, a_n$ of $Y(s)$ except $s = i\omega$ are in the left half-plane.

$$y(t) = \frac{1}{2\pi i} \lim_{T\to\infty} \int_{-iT}^{iT} \frac{H(s)e^{st}}{s - i\omega} \,\mathrm{d}s$$

Since $H(s)$ is meromorphic, by residue theorem,

$$y(t) = \operatorname{Res}(Y, i\omega) + \sum_{k=1}^{n} \operatorname{Res}\left(Y, a_k\right)$$

$$\operatorname{Res}(Y, i\omega) = \lim_{s\to i\omega} (s - i\omega)\frac{H(s)e^{st}}{s - i\omega} = H(i\omega)e^{i\omega t}$$

The other residues contribute multiple of $e^{-a_k t}$, which vanish as $t \to \infty$.

Thus, the steady response is

$$H(i\omega)e^{i\omega t}$$

$H(i\omega)$ is called frequency response, $|H(i\omega)|$ is called magnitude response, and $\operatorname{Arg}(H(i\omega))$ is called phase (angle) response.

x Low-pass (低通) (LP), high-cut, or bass-cut filter (LPF)

A continuous-time linear time-invariant system is a low-pass filter iff $H(i\omega)$ where $i = \sqrt{-1}$ as a function of a real variable ω is such that

$$\lim_{\omega\to 0} |H(i\omega)| = K \in \mathbb{R}_{>0}$$

and

$$\lim_{\omega\to\infty} |H(i\omega)| = 0,$$

where K is called low-pass gain. The filter is called prototype or canonical if $K = 1$.

xi High-pass (高通) (HP), low-cut, or treble-cut filter (HPF)

A continuous-time linear time-invariant system is a high-pass filter iff $H(i\omega)$ where $i = \sqrt{-1}$ as a function of a real variable ω is such that

$$\lim_{\omega\to 0} |H(i\omega)| = 0$$

and

$$\lim_{\omega\to\infty} |H(i\omega)| = K \in \mathbb{R}_{>0},$$

where K is called high-pass gain. The filter is called prototype or canonical if $K = 1$.

xii Band-pass (BP) filter (BPF)

A continuous-time linear time-invariant system is a band-pass filter iff $H(i\omega)$ where $i = \sqrt{-1}$ as a function of a real variable ω is such that

$$\lim_{\omega\to 0} |H(i\omega)| = 0$$

$$\lim_{\omega \to \infty} |H(i\omega)| = 0$$

and

$$\max_{\omega} |H(i\omega)| = K \in \mathbb{R}_{>0},$$

where K is called band-pass gain. The filter is called prototype or canonical if $K = 1$.

xiii Pole–zero plot or pole–zero constellation

A pole–zero plot or pole–zero constellation is a graphical representation showing poles and zeros of a transfer function in the complex plane.